NEW CHINA EQUITY MODEL LAUNCH SERIES

Shenzhen

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OVERVIEW OF THE NEW CHINA EQUITY MODEL

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Executive Director l APAC Equity Solutions Research

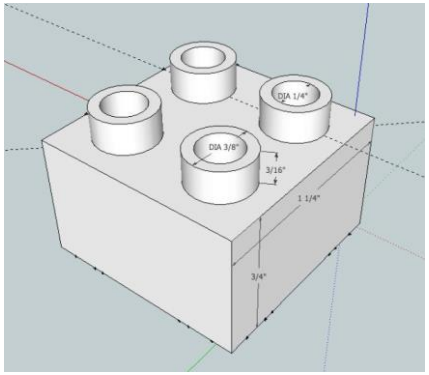
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FACTORS: A BRIEF REVIEW

Factors are systematic themes that influence asset returns and risks

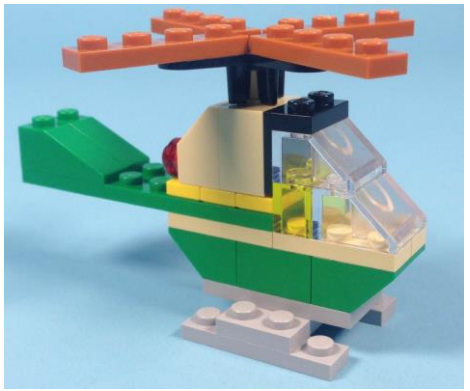
Macroeconomic Factors



Barra Factors
(Cross-sectional characteristics)



Factor Strategies
(e.g. MSCI China A High Dividend Yield)



Not investable

Used for asset allocation, modelling and understanding investment decisions

Used for communicating about investment decisions, as building blocks of strategies

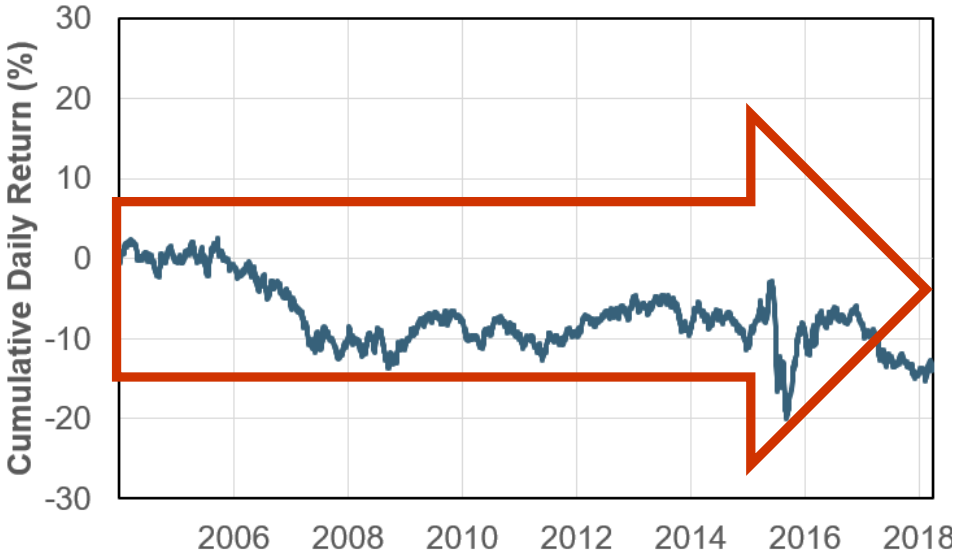
Investable

Used for implementing investment decisions, unbundling sources of return

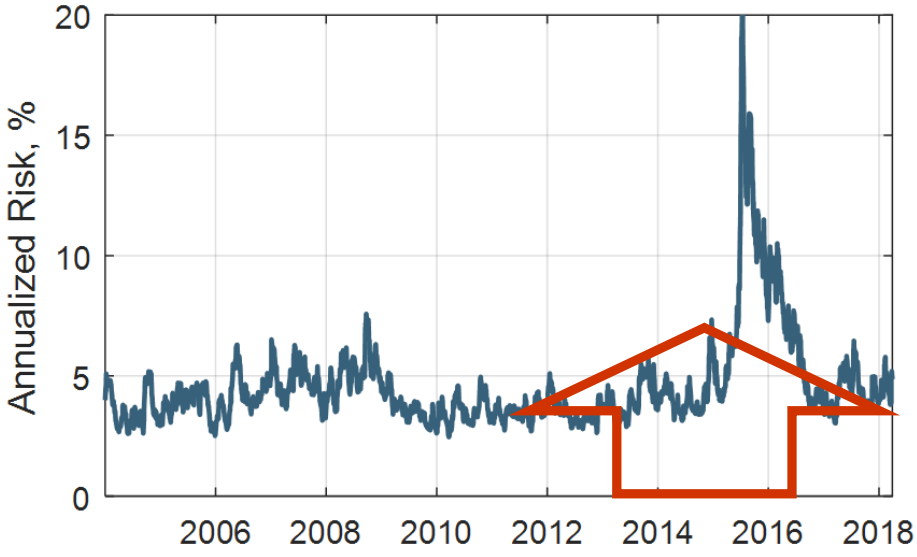
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WHAT IS A RISK FACTOR?

No persistent trend in return



High risk

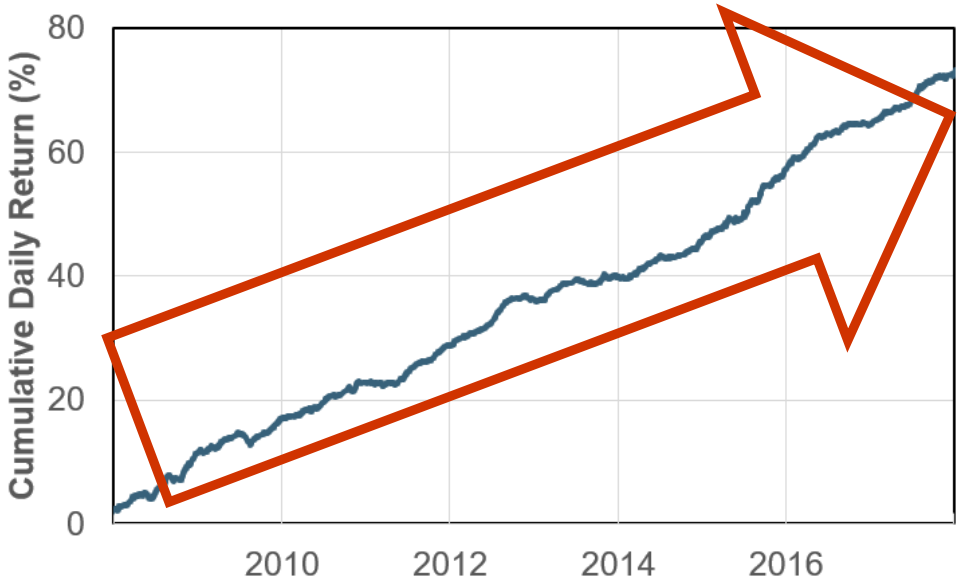


- Risk factors help forecast portfolio risk
- Using factor models in portfolio construction helps control portfolio volatility and fine-tune the sources of volatility

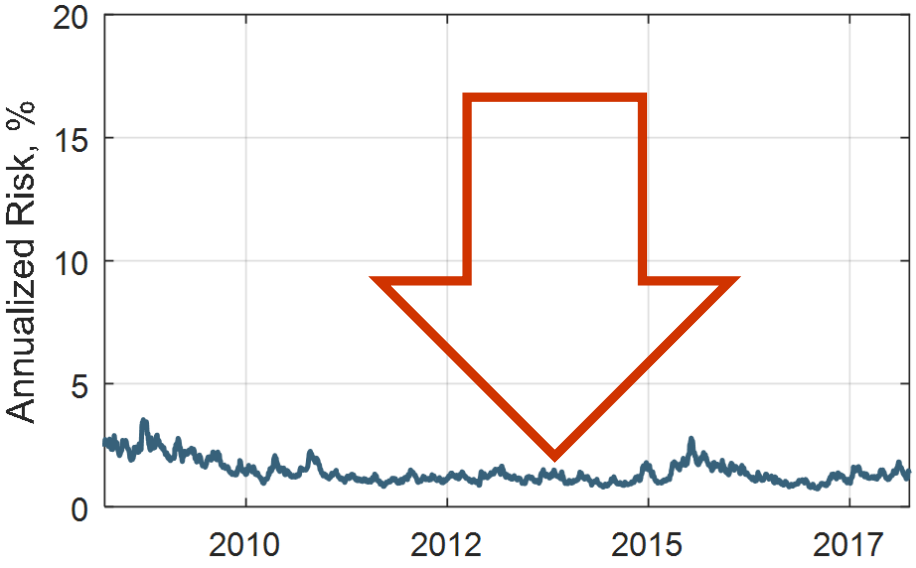
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WHAT ABOUT AN ALPHA FACTOR?

Strongly trending return



Low risk



- Alpha factors identify systematic characteristics that lead to outperformance
- Many factors share characteristics of both alpha and risk factors (we refer to these as systematic equity strategies [SES])

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WHY ARE SOME FACTORS REWARDED?

Behavioural Biases



Institutional Constraints



Underlying Risks

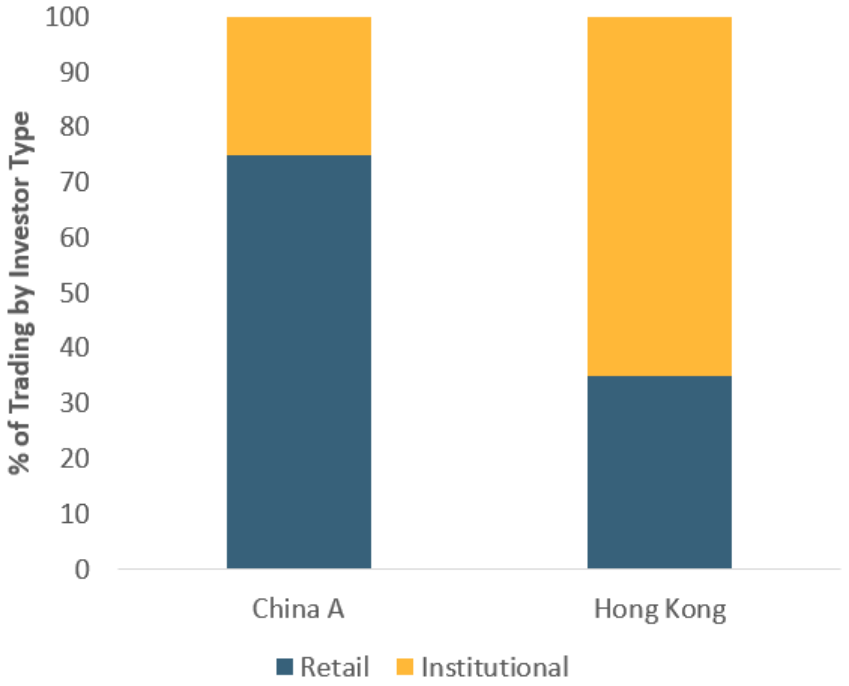


- Investor behavior, institutional framework and risk preferences are fundamental characteristics of economies and markets that change slowly (if at all) through time
- … Therefore, factor premia explained by these effects cannot be arbitraged away just because more investors follow the same strategies
- These effects exist through different markets, but their relative importance may be different depending on typical investor behaviour, regulations, the extent of institutional participation and the macroeconomic environment in different countries. This leads to differences in factor behaviour between markets

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WHY ARE CHINA A –SHARES SUITED TO QUANTITATIVE AND QUANTAMENTAL INVESTING?





High and rapidly changing market volatility highlights need for rigorous risk control

Low institutional participation and large proportion of retail investors creates opportunities to harvest alpha from behavioural mistakes

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MORE REASONS FOR FACTOR INVESTING IN A SHARES

China's retail investors create playground for data-loving quants

Five Reasons Why Global Investors Should Be Investing In China A-Shares

Quants Feast in the Land of Retail Investors

$103 Billion Quant Firm Says China Stocks `Very Compelling'

- Large breadth, large depth
- Most institutions are absolute return focused, high pressure of short-term ranking, high turnover of managers, etc.
- Government entities are trading with multiple goals (not for profit only)

MSCI EXPERTISE IN CHINA A-SHARES MODELLING

"By 1976 Barra (now part of MSCI) had created sophisticated models that predicted stock returns based on many different risk factors"

Bionic Beta Wins the 1970s
(Forbes, February 2014)

Modelling China's Equity Markets

- 2003: First generation model for A-shares
- 2012: Previous generation model for A-shares (CNE5)
- 2014: China International Model
  (Hong Kong Model built in 2003)
- 2017: All China Model
- 2018: New China A Total Market Equity Model and Trading Model
  China Integrated Model

KEY REASONS FOR BUILDING THE NEXT-GENERATION CHINA EQUITY MODEL

- **Changing structure of the A-shares market**
  - Greater internationalization/institutional participation suggested we should evaluate the effectiveness of new factors

- **Research innovations in latest generation Global, Asia and Emerging Market models**
  - SES factors: risk factors that are also generic sources of alpha
  - Alignment of factor decay and model horizon
  - Point-in-time fundamental data
  - Optimizing the size of the estimation universe

CNE5 FACTOR STRUCTURE

 Volatility	 Yield	 Quality	 Momentum	 Value	 Size	 Growth	 Liquidity	 Sentiment
Beta		Leverage	Momentum	Book-to-Price	Size	Growth	Liquidity	
Residual Volatility				Earnings Yield	Non-linear Size			

NEW CHINA EQUITY MODEL FACTOR STRUCTURE

Volatility	Yield	Quality	Momentum	Value	Size	Growth	Liquidity	Sentiment
Beta	**Dividend Yield**	Leverage	Momentum	Book-to-Price	Size	Growth	Liquidity	**Analysts Sentiment**
Residual Volatility		**Profitability**	**Short-Term Reversal**	Earnings Yield	Mid-Cap			~~**News Sentiment**~~
		Earnings Variability	**Industry Momentum**	**Long-Term Reversal**				~~**Short Interest**~~
		Earnings Quality	**Seasonality**					~~*Foreign Sensitivity*~~
		Investment Quality						~~*Dual Listing*~~

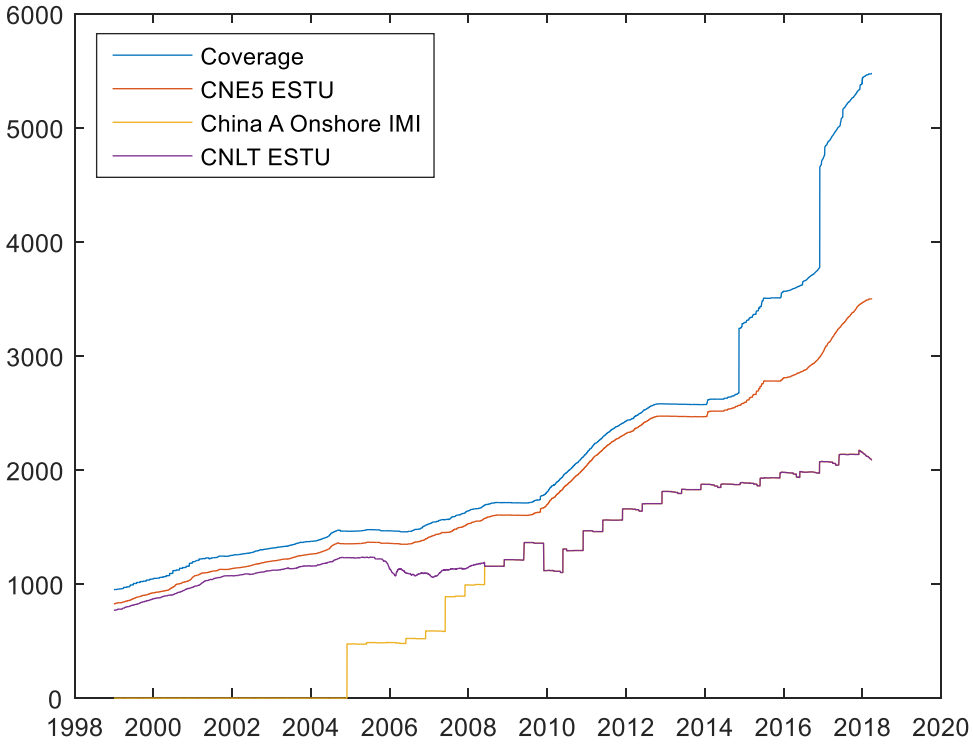
Factor names in **bold** indicate new factors in the long-term model

Factor names in **red** indicate additional new factors for the Trading Model

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ESTIMATION UNIVERSE

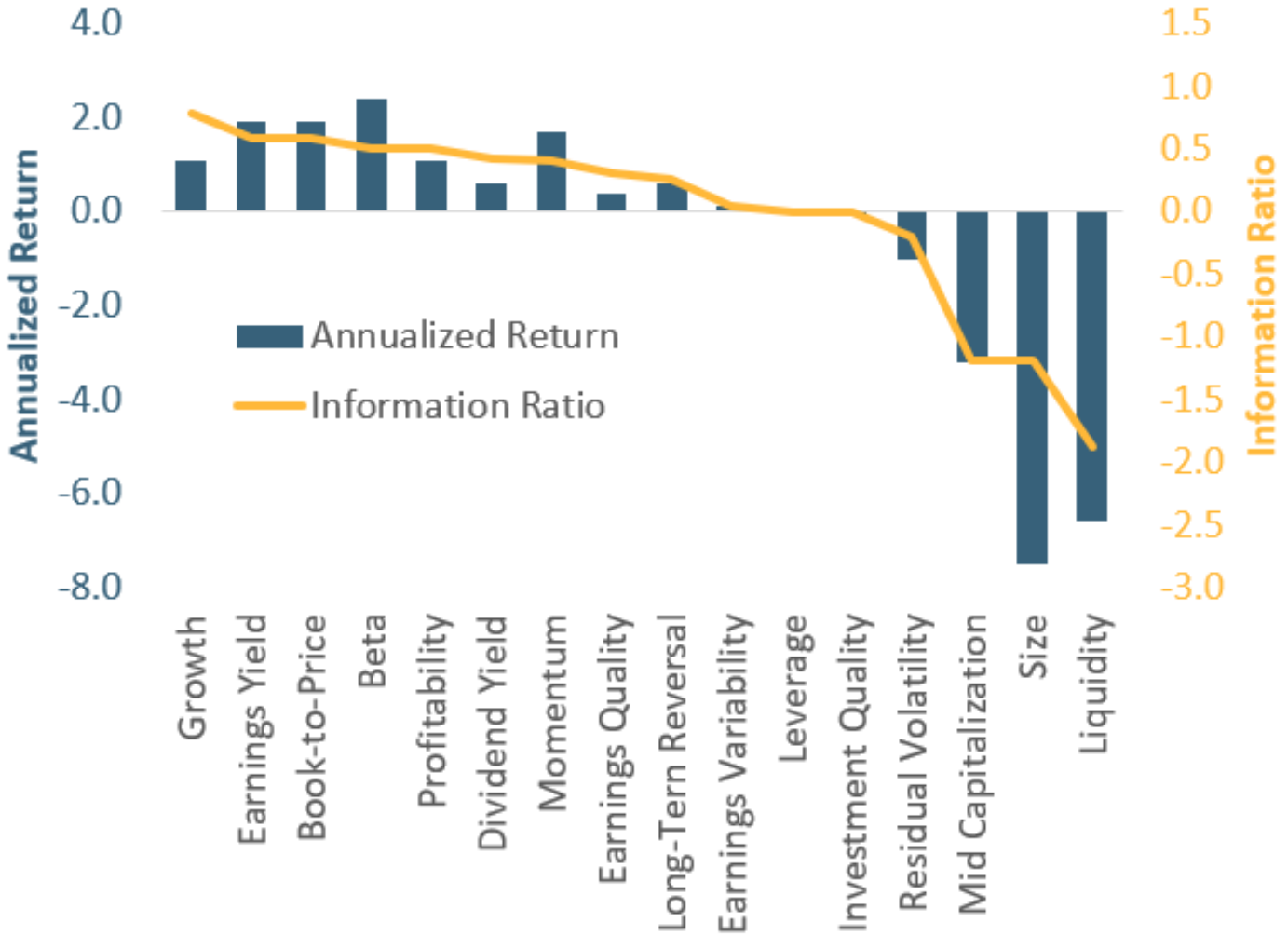
- **CNE5 Estimation Universe includes most stocks, except cross listings**
- **CNLT Estimation Universe**
  - Since June 2008: use MSCI China A Onshore IMI
  - Before June 2008: construct the Estimation Universe by following current MSCI ACWI IMI methodology, with proper size and liquidity thresholds



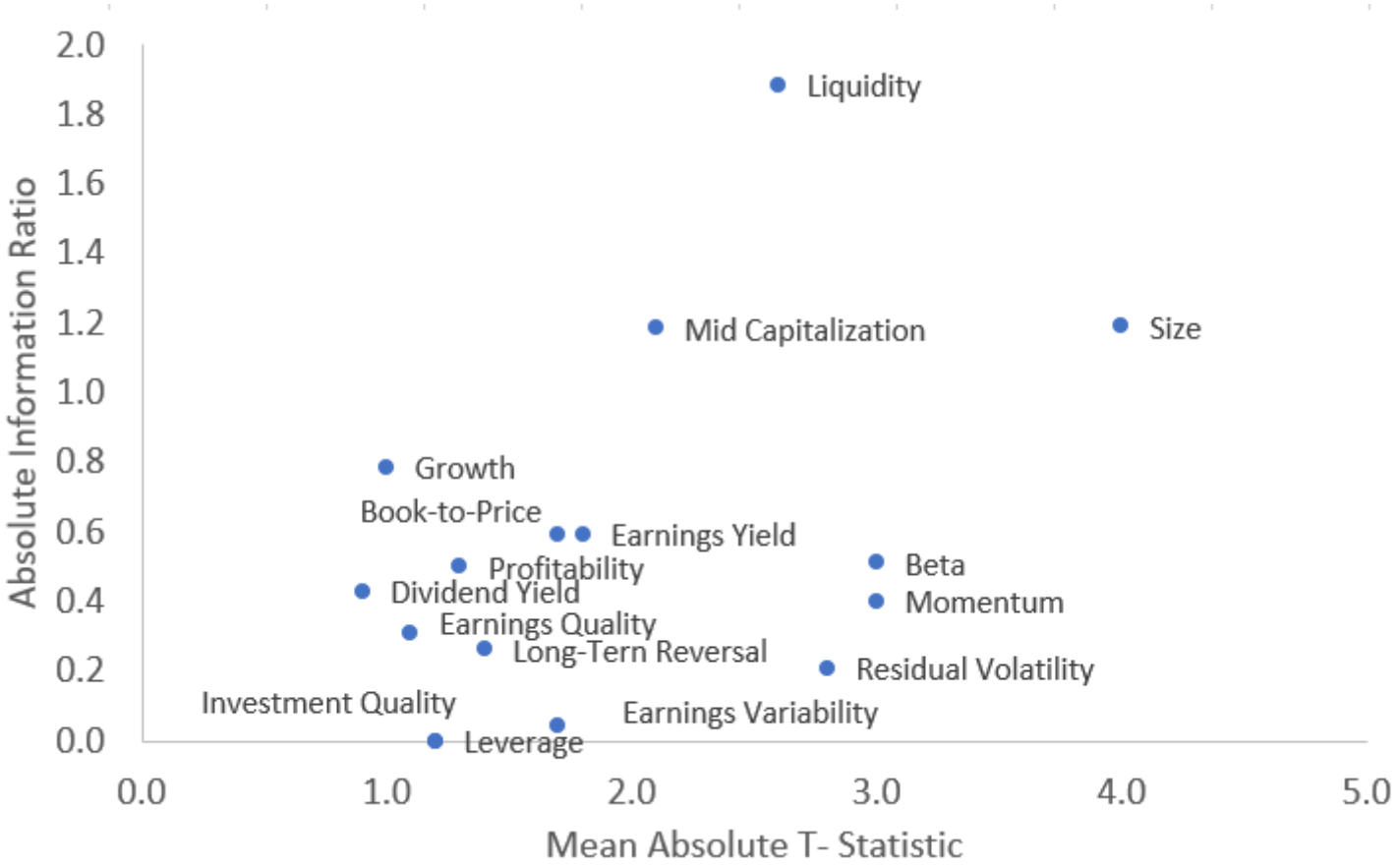
MONTHLY STATISTICS OF SLOW STYLE FACTORS

Factor	Mean |t|	|t|>2 %	Annual Return (%)	Annual Volatility (%)	Information Ratio	CV R2 Gain (bp)	Stud R2 Gain (bp)	Maximum drawdown (%)	Correlation with market	Volatility Inflation Factor
Size	4.0	70.2	-7.5	6.3	-1.2	80.4	82.6	117.7	0.0	6.41
Momentum	3.0	57.3	1.7	4.9	0.4	44.7	48.2	15.8	-0.1	2.73
Residual Volatility	2.8	57.3	-1.0	4.8	-0.2	28.7	33.1	24.7	0.3	2.7
Beta	3.0	57.9	2.4	4.7	0.5	39.8	43.4	8.0	0.5	2.48
Liquidity	2.6	53.2	-6.6	3.5	-1.9	27.0	31.5	97.1	0.3	4.62
Earnings Yield	1.8	29.2	1.9	3.2	0.6	9.5	13.5	7.2	0.3	4.62
Book-to-Price Ratio	1.7	34.5	1.9	3.2	0.6	11.2	14.9	7.5	0.1	3.31
Mid Capitalization	2.1	44.4	-3.2	2.7	-1.2	20.2	23.6	49.4	0.1	1.39
Long-Term Reversal	1.4	24.6	0.6	2.3	0.3	3.9	8.8	7.1	0.2	2.06
Earnings Variability	1.7	35.0	0.1	2.3	0.0	6.4	11.0	6.6	0.2	1.46
Profitability	1.3	17.0	1.1	2.2	0.5	0.3	5.4	8.4	-0.2	3.15
Leverage	1.2	20.5	0.0	1.9	0.0	1.5	5.7	9.0	0.0	2.09
Investment Quality	1.2	19.6	-0.1	1.8	0.0	-0.3	4.6	5.8	0.1	1.44
Dividend Yield	0.9	9.9	0.6	1.4	0.4	-3.1	1.3	5.3	0.0	2.86
Growth	1.0	12.9	1.1	1.4	0.8	-3.8	1.3	2.4	-0.2	1.66
Earnings Quality	1.1	15.8	0.4	1.3	0.3	-2.1	3.0	3.6	0.0	1.24

LONG TERM MODEL: KEY STYLE FACTOR CHARACTERISTICS

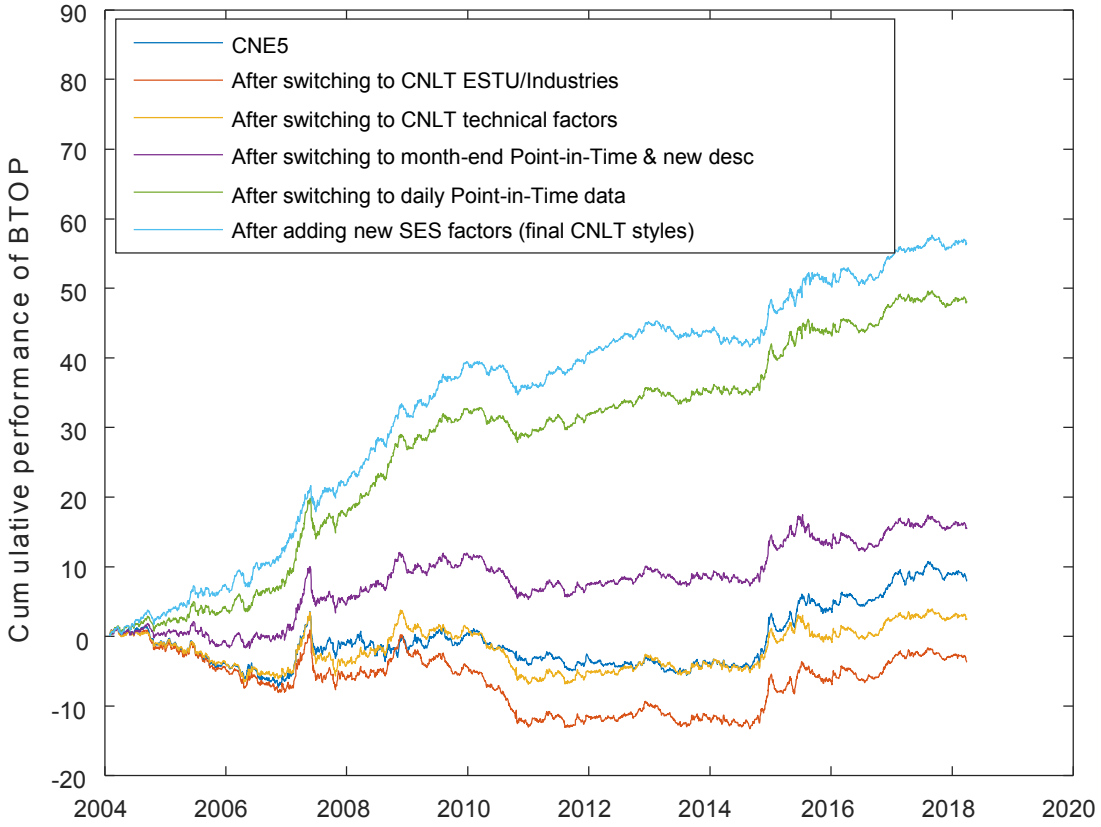


LONG TERM MODEL: RISK AND ALPHA PROPERTIES OF STYLE FACTORS

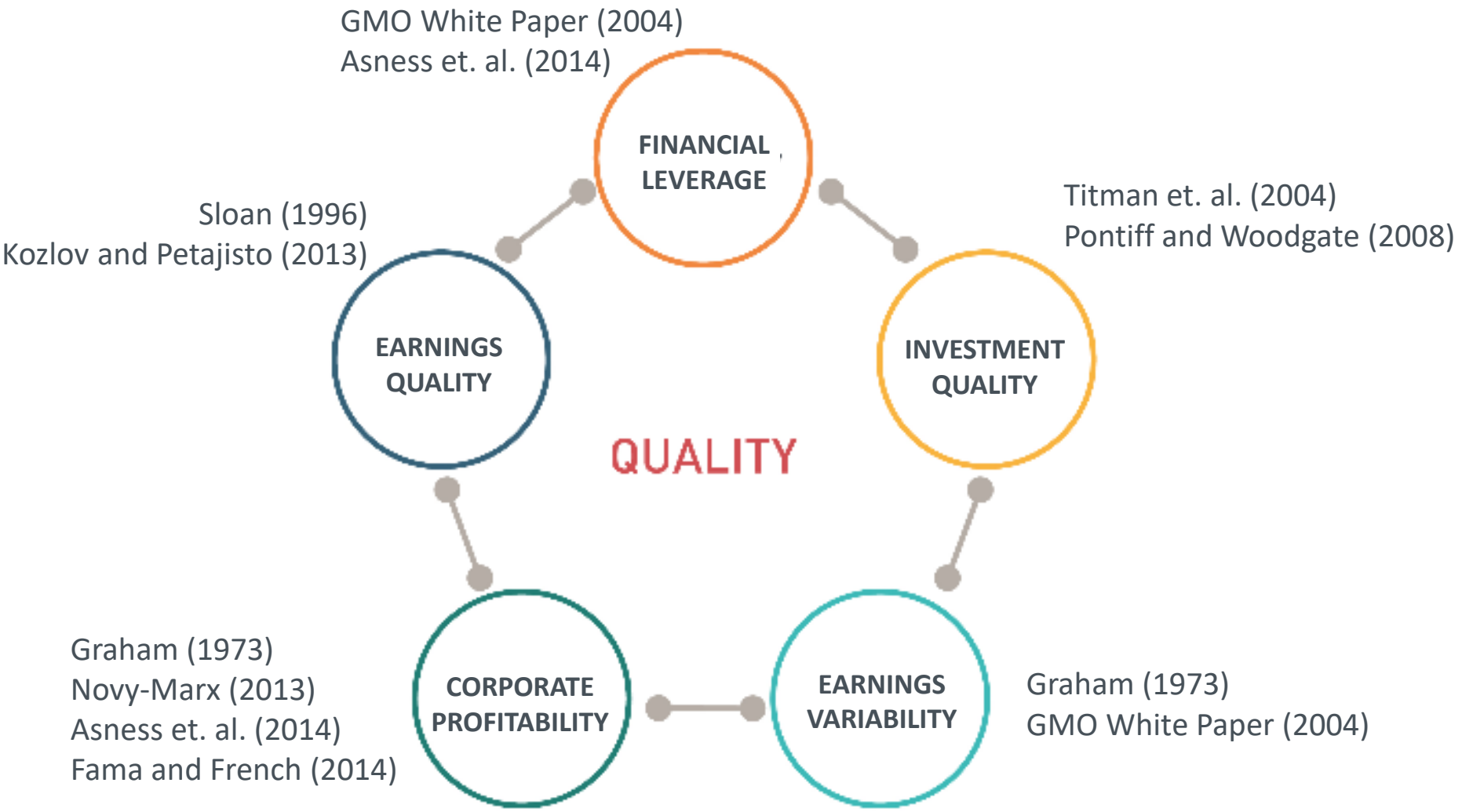


PERFORMANCE OF BOOK-TO-PRICE: POINT-IN-TIME DATA

- **What drives the difference between CNE5 and CNLT**
  - The daily updates of point-in-time data

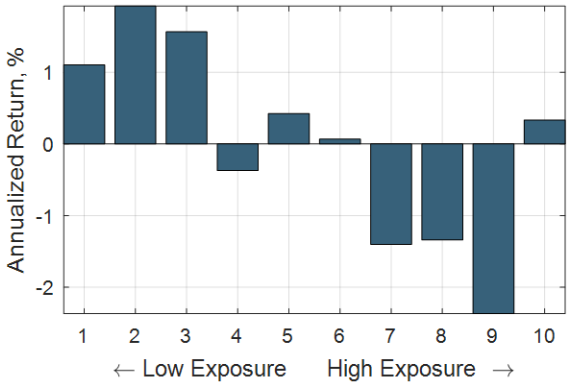


QUALITY: A MULTI-DIMENSIONAL INVESTMENT CONCEPT FAVORED BY INSTITUTIONAL INVESTORS

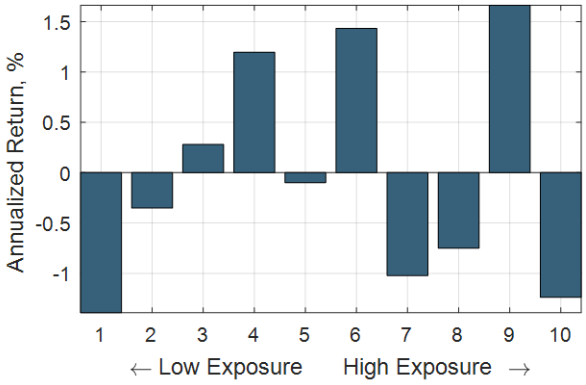


PERFORMANCE OF QUALITY IN CHINA A SHARES

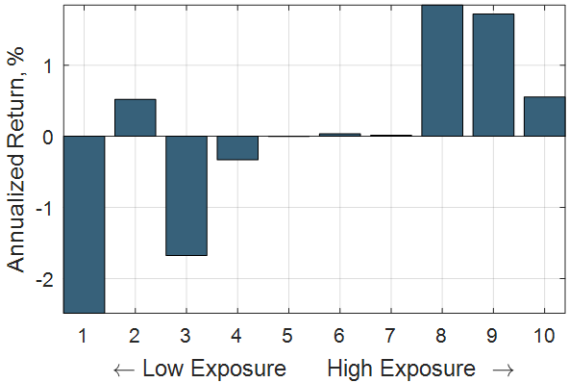
Leverage



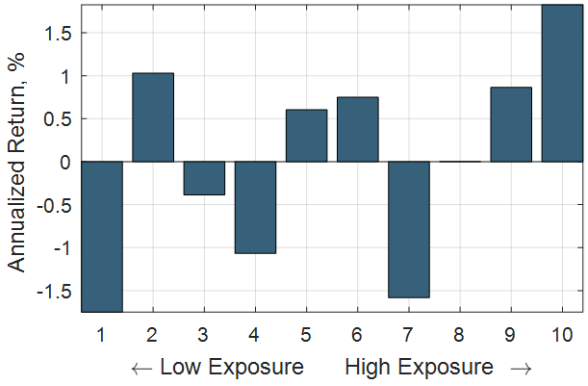
Earnings Quality



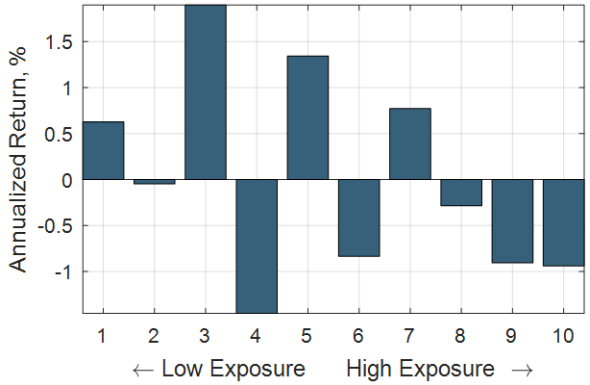
Investment Quality



Profitability



Earnings Variability



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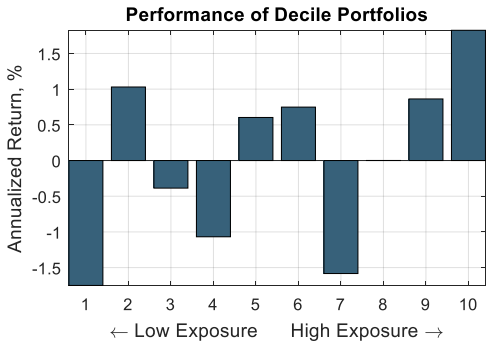
RISK AND ALPHA DIMENSIONS OF QUALITY

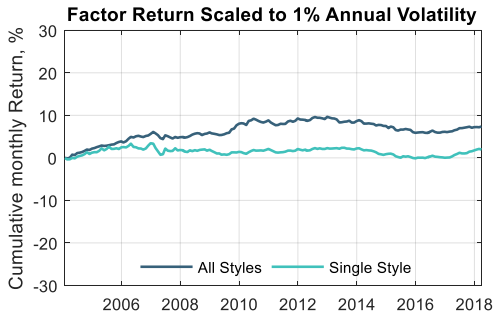
	Annual Return (%)	Annual Risk (%)	IR	Mean absolute T-stat	Absolute T-stat >2, %
Profitability	1.1	2.2	0.5	1.3	17.0
Earnings Quality	0.4	1.3	0.3	1.1	15.8
Earnings Variability	0.1	2.3	0.0	1.7	35.0
Leverage	0.0	1.9	0.0	1.2	20.5
Investment Quality	-0.1	1.8	0.0	1.2	19.6

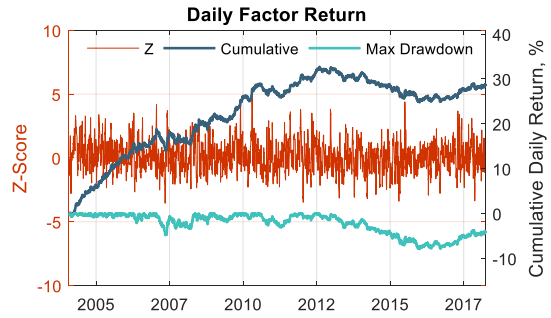
FACTOR DETAILS: PROFITABILITY

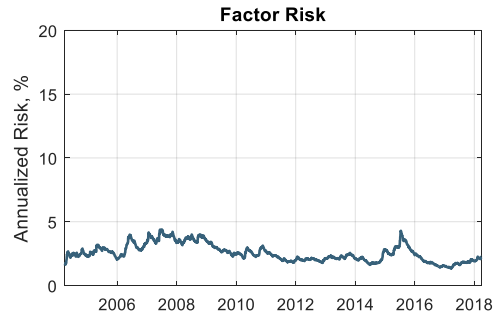
- **Measures the efficiency of a firm's operations and total activities.**

- **Descriptors**
  - Asset Turnover
  - Gross Profitability
  - Gross Profit Margin
  - Return on Assets







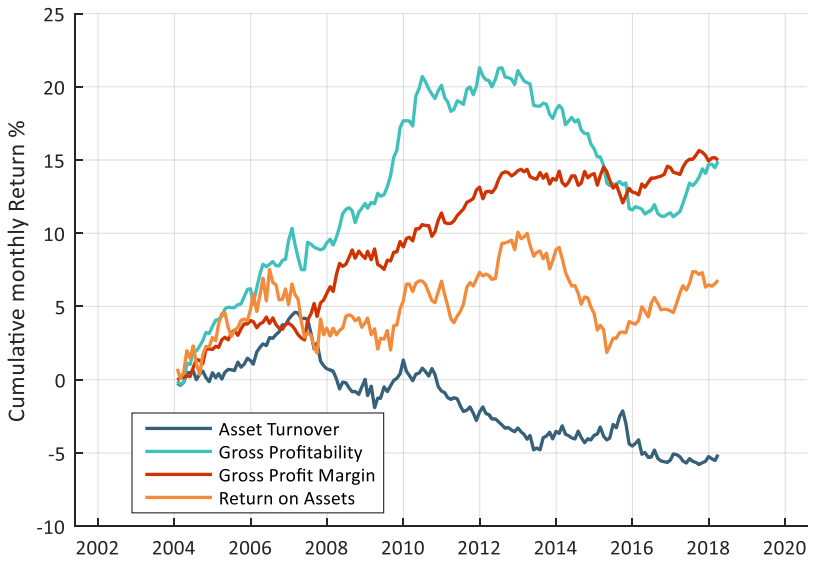


Monthly regression statistics

	Mean |t|	|t|>2 (%)	Annual Return (%)	Annual Volatility (%)	Information Ratio	CV R2 Gain (bp)	Stud R2 Gain (bp)	Maximum drawdown (%)	Correlation with market
Asset Turnover	1.1	15.8	-0.4	1.5	-0.2	-2.1	2.8	10.4	-0.14
Gross Profitability	1.2	21.6	1.0	2.0	0.5	0.4	5.4	10.2	-0.24
Gross Profit Margin	1.1	16.4	1.1	1.5	0.7	-2.4	2.5	2.5	-0.07
Return on Assets	1.3	23.4	0.5	2.5	0.2	0.0	5.2	8.2	-0.07
Profitability	1.3	17.0	1.1	2.2	0.5	0.3	5.4	8.4	-0.23

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PROFITABILITY DESCRIPTORS



Time Series Correlation of Returns

		1	2	3	4	5
1	Asset Turnover	1				
2	Gross Profitability	0.41	1			
3	Gross Profit Margin	-0.56	0.31	1		
4	Return on Assets	-0.033	0.35	0.34	1	
5	Profitability	0.37	0.9	0.44	0.63	1

Average Cross-Sectional Correlation

		1	2	3	4	5
1	Asset Turnover	1				
2	Gross Profitability	0.46	1			
3	Gross Profit Margin	-0.51	0.31	1		
4	Return on Assets	0.19	0.58	0.34	1	
5	Profitability	0.46	0.9	0.45	0.81	1

EXPLANATORY POWER

Estimate incremental models to gauge the impact of innovations on the use cases and results
New SES factors are impactful in the explanatory power of the model

ESTU	Industries	Styles	Monthly CV R2	Monthly Studentized R2
CNLT ESTU	**CNLT Industries**	**CNE5 Styles (**Beta, Book-to-Price, Earnings Yield, Growth, Leverage, Liquidity, Mid Cap, Momentum, Residual Volatility, Size**)**	0.4008	0.4206
CNLT ESTU	**CNLT Industries**	**Replace CNE5 styles with the corresponding CNLT styles** (including Point-In-Time Fundamental data)	0.4017	0.4216
CNLT ESTU	**CNLT Industries**	**Add new slow SES factors (**Earnings Quality, Long-Term Reversal, Profitability, Dividend Yield, Investment Quality**) plus Earnings Variability**	0.4026	0.4235

Point-in-Time data and the innovations in the existing styles improve R2

The new SES factors improve R2

*Studentized R2 is similar to adjusted R2, but for each asset i, the residual from the regression is divided by $\sqrt{1-\left(W^{\frac{1}{2}}X(X'WX)^{-1}X'W^{\frac{1}{2}}\right)_{ii}}$, where X is the exposure matrix, and W is regression weighting scheme.

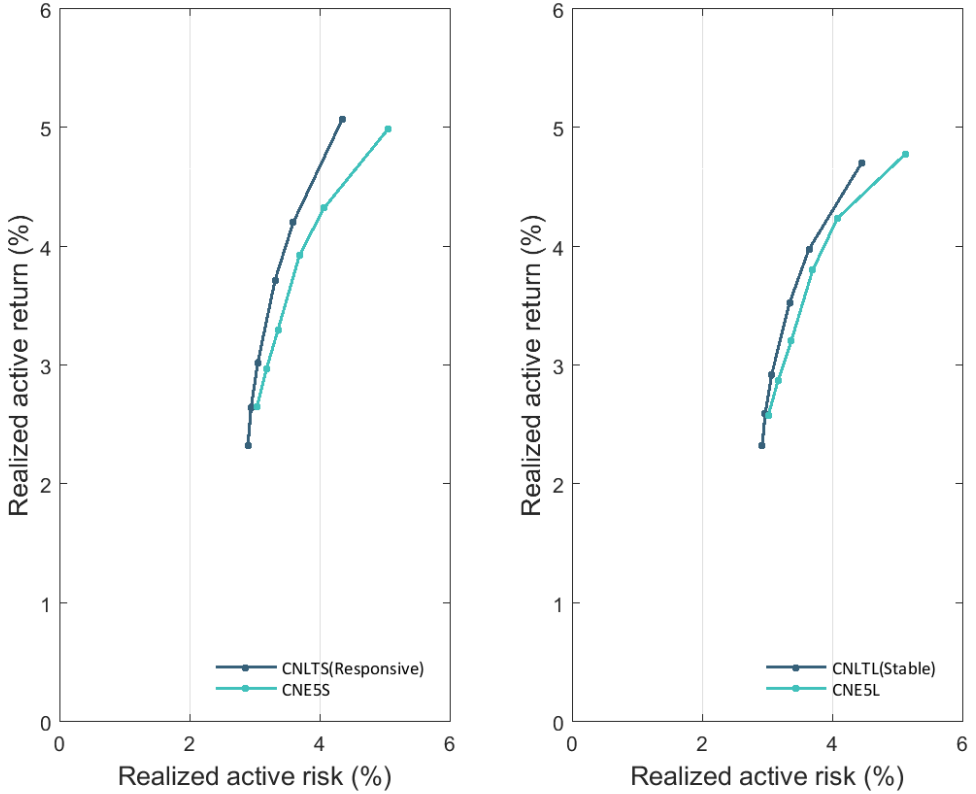
PORTFOLIO RISK FORECAST ACCURACY

	Portfolio	Turnover	Risk	Bias				Q			
				CNLTS	CNLTL	CNE5S	CNE5L	CNLTS	CNLTL	CNE5S	CNE5L
Long only	Random	52.04	31.90	1.06	1.02	1.15	1.09	2.4675	2.5162	2.5119	2.5414
	CNLT Industry Tilt	55.62	35.81	1.05	1.02	1.13	1.10	2.4792	2.5187	2.5210	2.5531
	CNE5 Industry Tilt	55.93	35.99	1.05	1.03	1.13	1.10	2.4849	2.5297	2.5236	2.5637
	CNLT Style Tilt	176.34	31.99	1.07	1.03	1.16	1.11	2.5001	2.5279	2.5445	2.5643
	CNE5 Style Tilt	206.58	31.84	1.08	1.03	1.16	1.11	2.5103	2.5380	2.5557	2.5696
Long/short	Random Active	52.04	10.71	1.01	1.01	1.06	1.04	2.4767	2.5168	2.4951	2.5366
	CNLT Industry Tilt Active	55.62	20.33	1.08	1.08	1.16	1.14	2.5908	2.6391	2.6504	2.6962
	CNE5 Industry Tilt Active	55.93	20.42	1.08	1.08	1.15	1.13	2.5965	2.6356	2.6473	2.6917
	CNLT Style Tilt Active	176.34	12.33	1.06	1.07	1.17	1.16	2.5448	2.6140	2.6267	2.6959
	CNE5 Style Tilt Active	206.58	12.37	1.08	1.08	1.15	1.14	2.6211	2.6527	2.6713	2.7235
	CNLT Style Spread	137.95	13.28	1.08	1.06	1.31	1.29	2.3491	2.3973	2.5652	2.5818
	CNE5 Style Spread	180.47	14.39	1.11	1.10	1.16	1.14	2.4246	2.4629	2.4986	2.5182
	CNLTS Style Optimized	161.48	4.99	1.10	1.06	1.14	1.10	2.4763	2.4822	2.5039	2.4993
	CNLTL Style Optimized	154.72	4.97	1.09	1.06	1.13	1.11	2.4706	2.4785	2.4976	2.4948
	CNE5S Style Optimized	192.58	5.72	1.13	1.09	1.16	1.12	2.5113	2.5036	2.5376	2.5262
	CNE5L Style Optimized	188.89	5.70	1.12	1.09	1.16	1.13	2.4877	2.4878	2.5160	2.5105

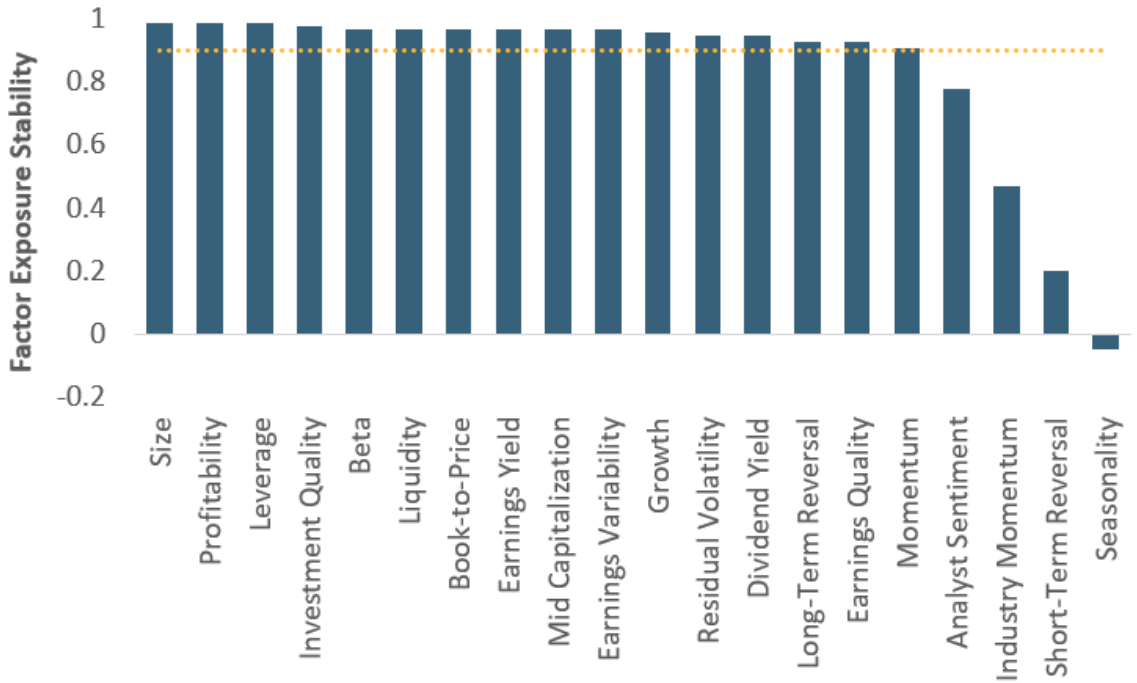
*CNLTS: CNLT responsive version
*CNLTL: CNLT stable version

BACKTEST ACTIVE STRATEGY

- **Alpha: weight Systematic Equity Strategy factors by IR**
- **Long only, 8% monthly turnover**
- **Benchmark & investment universe: CNLT Estimation Universe**



CONSTRUCTING A TRADING MODEL: FAST AND SLOW FACTORS



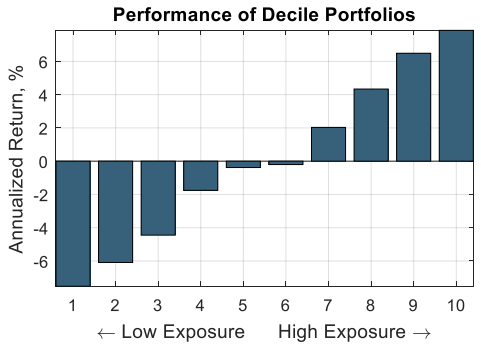
Factors below this stability threshold are only included in the daily model

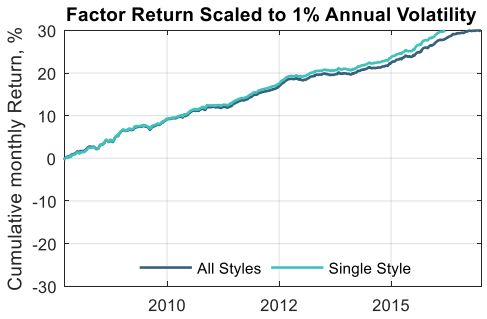
- Factor exposure stability is directly linked to portfolio turnover
- For faster factors, the changes in underlying information is faster and larger, implying that portfolio positions need to be rebalanced more frequently and more substantially to reflect the updates in factor exposures optimally

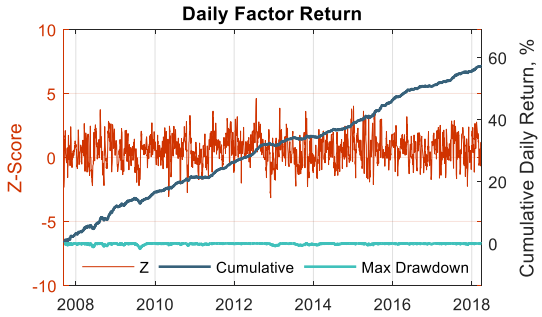
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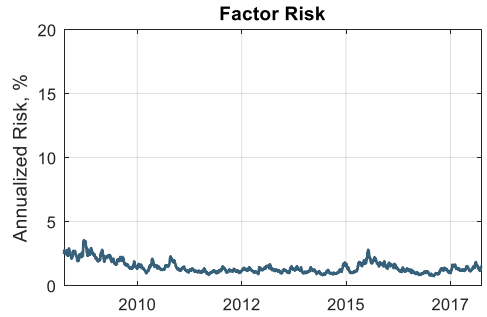
ANALYST SENTIMENT

- **Changes in analyst earnings forecast**
- **Descriptors**
  - Analyst revision ratio
  - Change of forecast E/P
  - Change of forecast earnings







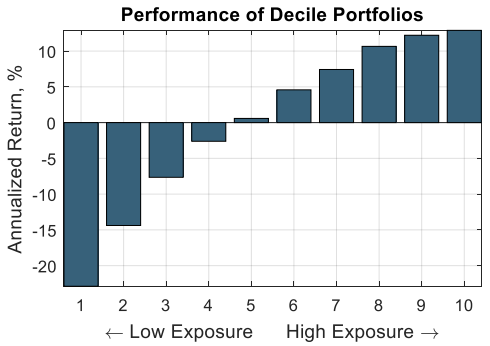


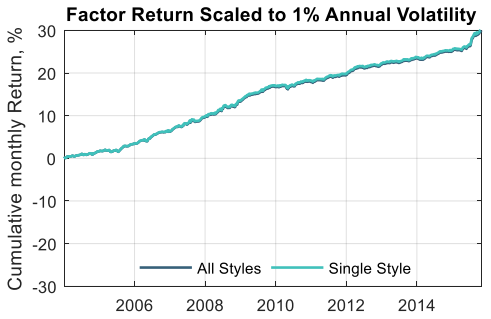
Weekly regression statistics

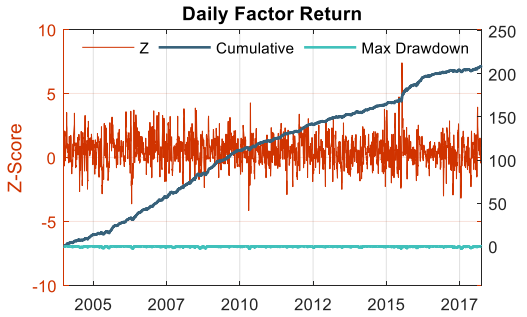
	Mean |t|	|t|>2 (%)	Annual Return (%)	Annual Volatility (%)	Information Ratio	CV R2 Gain (bp)	Stud R2 Gain (bp)	Maximum drawdown (%)	Correlation with market
Revision Ratio	1.2	16.3	5.2	1.6	3.3	-0.6	4.3	1.6	0
ETOP Forecast Change	1.3	19.6	3.8	1.8	2.2	-0.6	4.6	1.9	0.08
Earnings Forecast Change	1.3	22.2	4.2	1.7	2.5	-0.2	4.9	1.9	0.03
Analysts Sentiment	1.3	19.8	4.7	1.4	3.3	-0.1	5.0	1.5	0.02

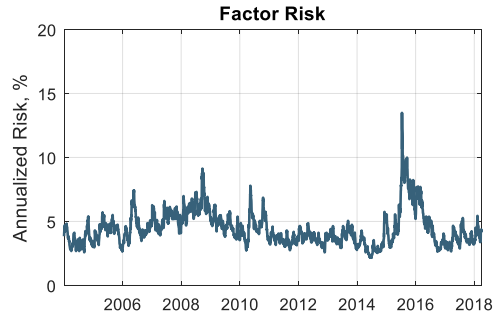
SHORT-TERM REVERSAL

- **Short-term price trend**
- **Descriptor**
  - Short-term reversal









Weekly regression statistics

	Mean |t|	|t|>2 (%)	Annual Return (%)	Annual Volatility (%)	Information Ratio	CV R2 Gain (bp)	Stud R2 Gain (bp)	Maximum drawdown (%)	Correlation with market
Short-Term Reversal	3.6	64.1	10.3	4.0	2.5	69.9	74.2	3.5	0.21

SUMMARY

- **The new China A Total Market Model**
  - Estimation Universe is created with Size/Liquidity/Suspension screening
  - Fundamental data is point in time with daily updates
  - Long term model has stable and responsive variants; new trading model has fast style factors
  - New slow decay factors reflect the growing institutionalization and internationalization of the China A-shares market, with focus on quality investing and fundamentals
  - Fast decay factors capture investor sentiment and behavioral patterns

MARKET INSIGHTS & PORTFOLIO MANAGEMENT WITH THE NEW CHINA EQUITY MODEL

新一代中国股票模型：市场洞察和投资组合管理

Andrew Tong

Executive Director l Client Coverage

Ader Xu

Vice President l Client Coverage

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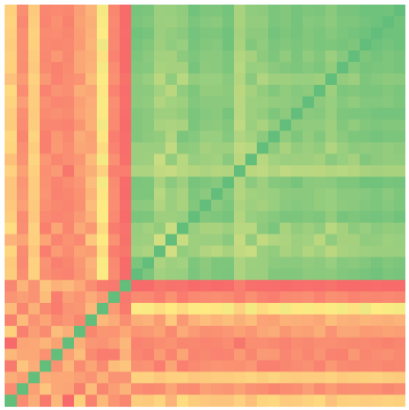
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AGENDA

- Overview
- Crowding: Risk or Return
- Factor framework for China A-shares market

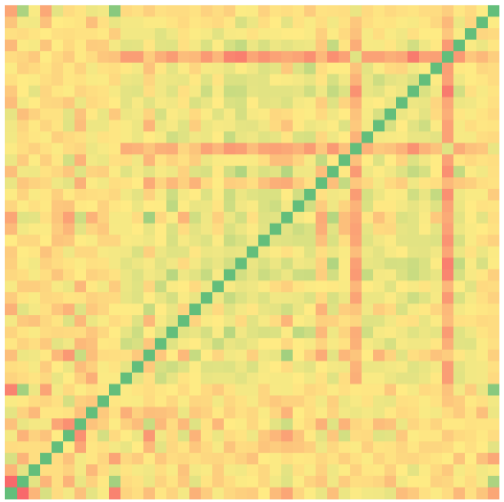
EVOLUTION OF BARRA CHINA EQUITY MODEL

CHE2



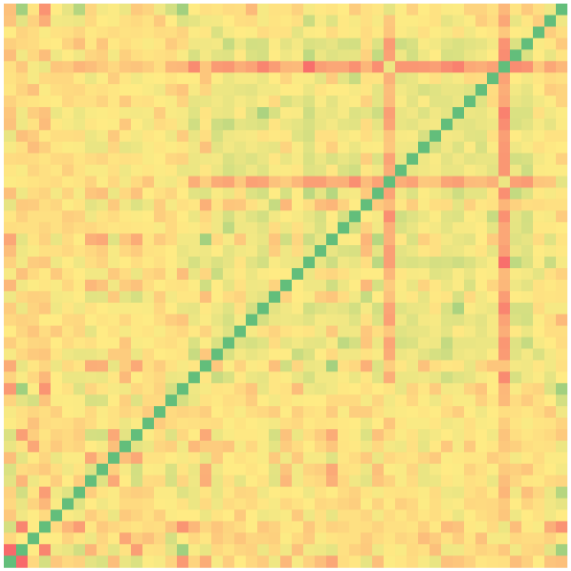
2005

CNE5



2012

CNLT



2018

1 0.8 0.6 0.4 0.2 0 -0.2 -0.4 -0.6 -0.8 -1

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Correlation Matrix Heat Map of Barra China Equity Models CHE2, CNE5, CNLT.
Date: 2018-07-31

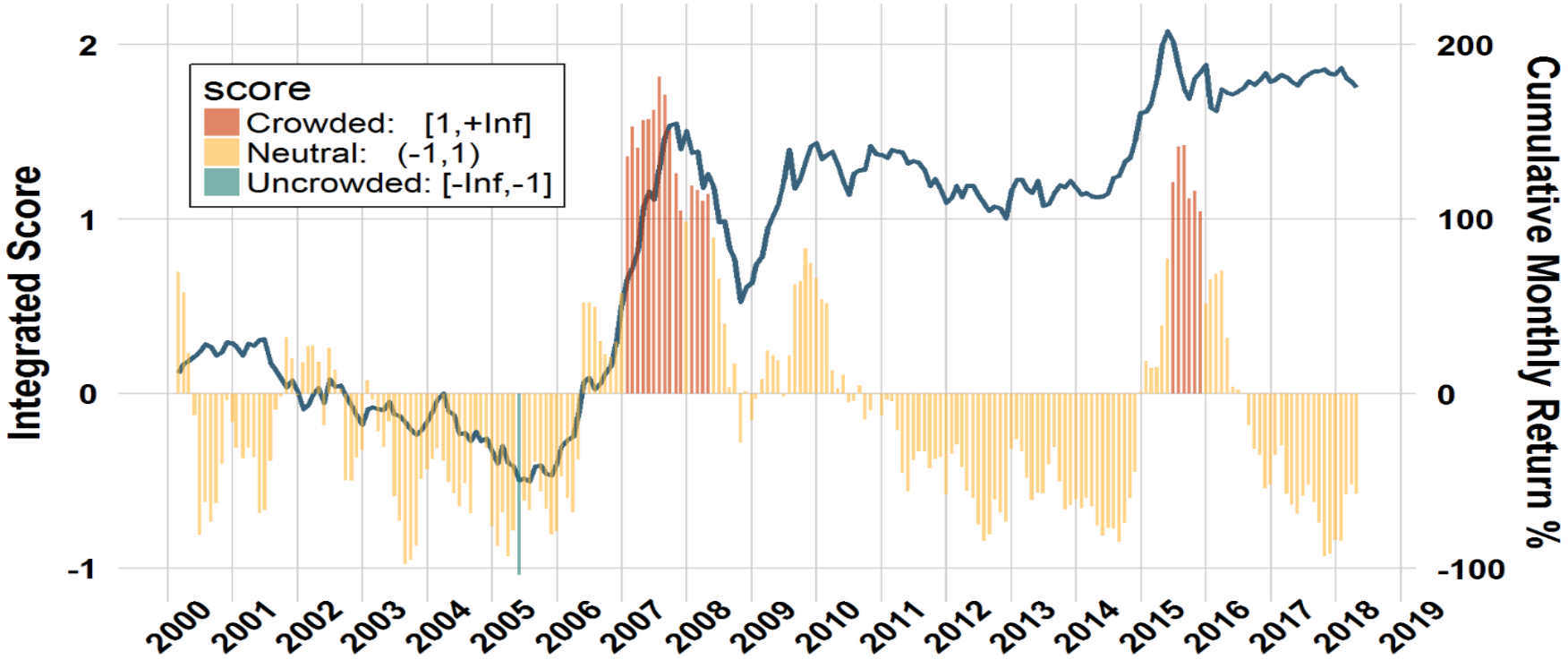
BARRA STYLE FACTOR STRUCTURE

VOLATILITY	QUALITY	MOMENTUM	VALUE	SIZE	GROWTH	LIQUIDITY	SENTIMENT
Beta ● Hist Beta **Residual Volatility** ● Hist Sigma ● Daily Std Dev ● Cumulative Range	**Leverage** ● Debt to Assets ● Book Leverage ● Mkt Leverage **Profitability** ● Asset Turnover ● Profitability ● Profit Margin ● Return on Assets **Earnings Variability** ● Variation in Sales ● Variation in Earnings ● Variation in Cash Flow ● Variation in Fwd EPS **Earnings Quality** ● Accruals - B/S Statement ● Accruals - C/F Statement	**Momentum** ● Rel Strength ● Hist Alpha **Industry Momentum** ● Industry Momentum **Short-Term Reversal** ● Short-Reversal **Seasonality** ● Quarterly Seasonality ● Annual Seasonality	**Book-to-Price** ● Book-to-price **Earnings Yield** ● Reported E/P ● Forecasted E/P ● Cash E/P ● EBIDTA/EV ● *Reported D/P* ● *Forecast D/P* **Long-Term Reversal** ● LT Rel Strength ● LT Hist Alpha	**Size** ● Log of Market Cap **Mid Cap** ● Cube of Size	**Growth** ● Sales Growth ● Earnings Growth ● Forecast LTG	**Liquidity** ● 1m Turnover ● 3m Turnover ● 12m Turnover ● 12m ATVR	**Analyst Sentiment** ● Revision Ratio ● Change in Analyst-Predicted E/P ● Change in Analyst-Predicted EPS

CROWDING: RISK OR RETURN?

CROWDING IN CHINA A SHARES MARKET

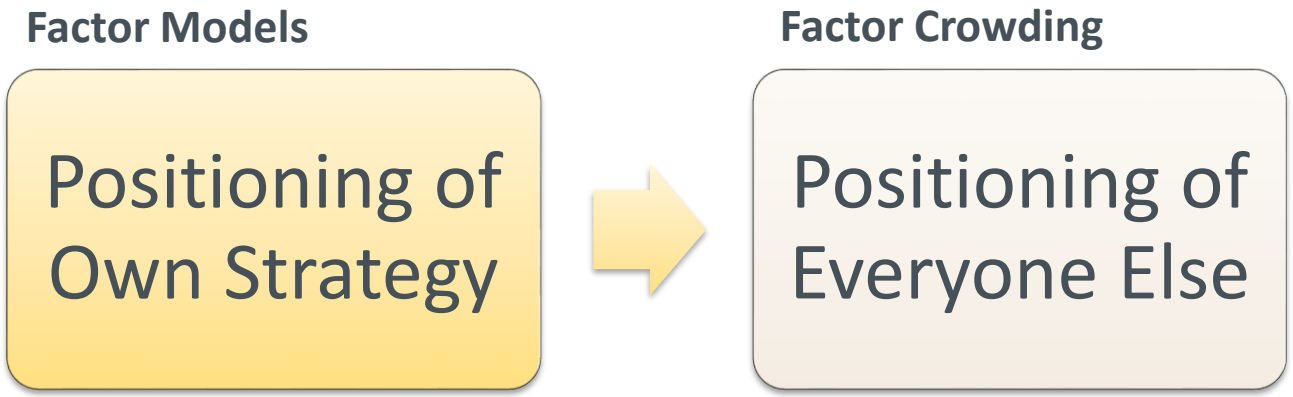
- Two notable crowded periods in recent history: Jan 2007 to April 2008, and Jun 2015 to Dec 2015.
- How to define crowded trade or strategy in the market?



WHY MEASURE CROWDING?

- Growth and Popularity of factors and factor investing increases potential for strategy crowding.
  - ➢ Risk of liquidity or drawdown event when positioning changes.

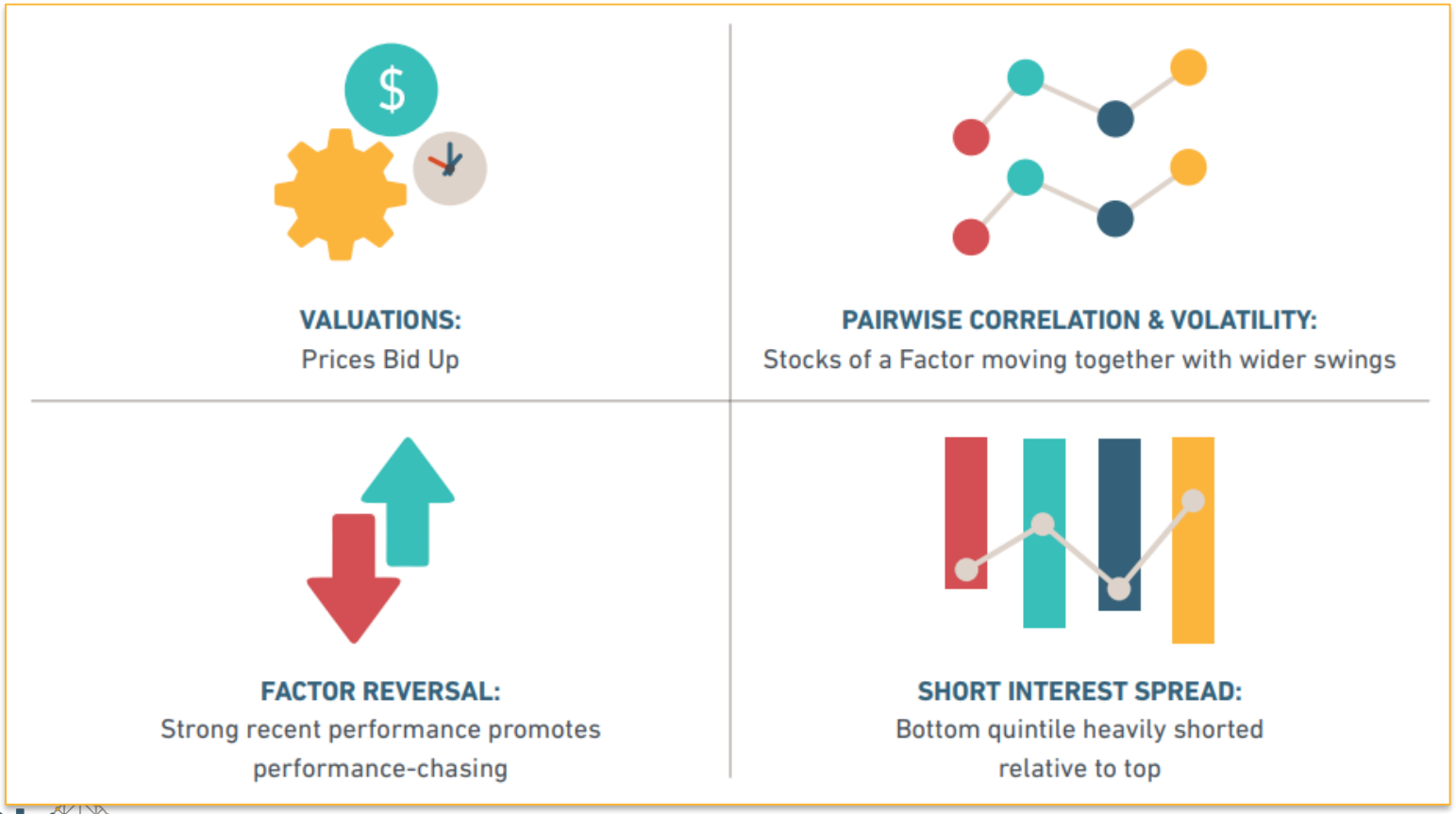
Factor crowding analyzes active managers' bets compared to those of the industry as a whole.



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THE MSCI FACTOR CROWDING MODEL

- **Crowding score** per factor
- **Additional crowding metrics**
- Daily history from **1996**
- Available for **US equity market**
- Available via **historical flat files**
- Global & asset level under research



MSCI CROWDING METRICS

1) **Valuation spreads -** Valuation of bottom vs. top quintile stocks, combining Book/P, Sales/P, forecasted E/P. Crowded factors will have expensive top quintile relative to bottom quintile.

2) **Short Interest spread** - Difference in Utilization ratio of bottom quintile stocks vs. top quintile, accounting for differences in shorting due to value, size and momentum factors. Crowded factors will have heavily shorted bottom quintile compared to top quintile.

3) **Pairwise correlation** - Average correlation of stocks in top and bottom quintiles of factor to each other over last 63 days, accounting for returns due to market, size, beta, residual volatility. Crowding will increase correlations.

4) **Factor Volatility** - Forecast factor volatility relative to forecast market volatility. Crowding will increase volatility.

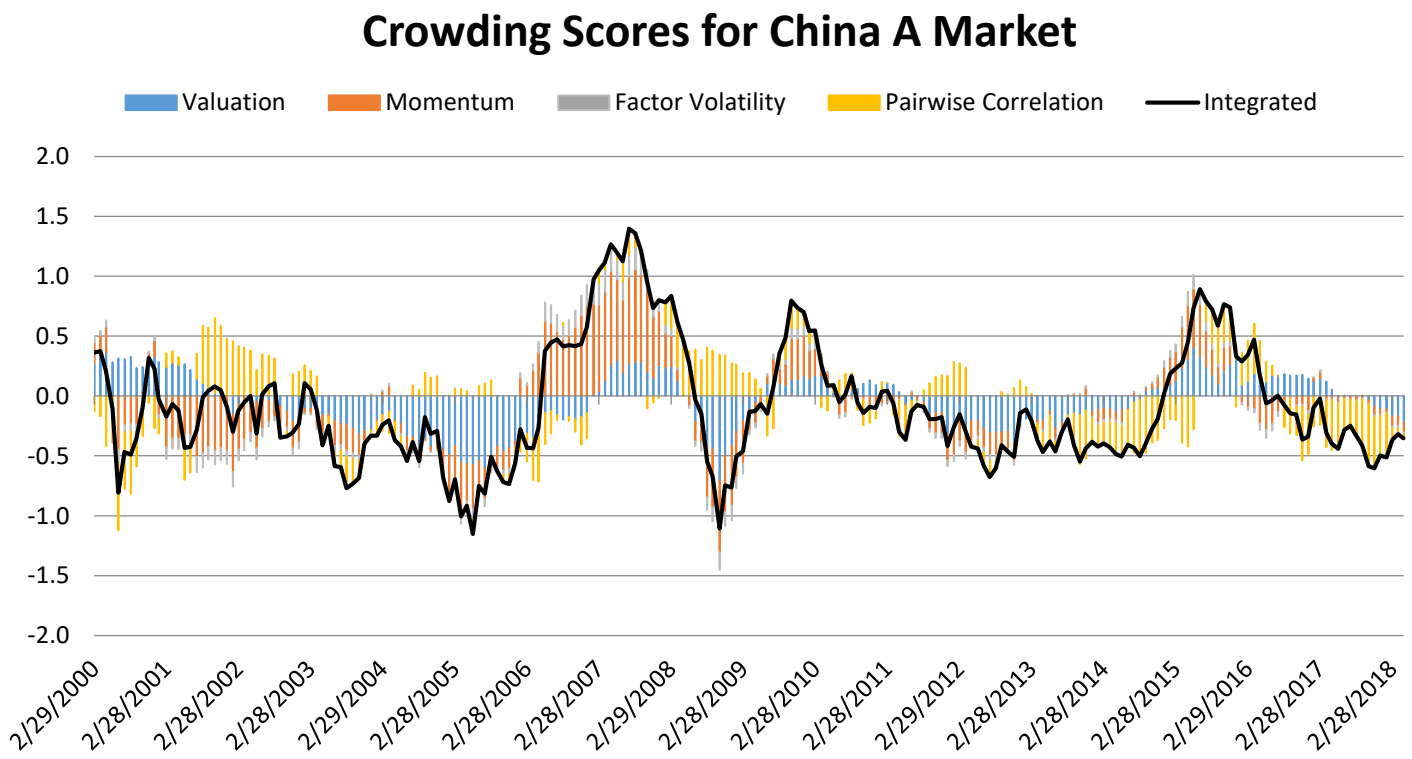
5) **Factor Reversal** - Cumulative trailing factor return over last 36 months. Strong trailing performance increases potential for crowding as investors chase prior performance.

6) **Integrated Score** - Single, overall measure of crowding, combination of #1-5.

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CROWDING IN CHINA A SHARES MARKET

- Momentum and Valuation decay faster than Factor Volatility score.
- Pairwise correlation tends to be higher when the market plunges.



Crowding Scores for China A Market and factors are computed with CNLTS data according to the MSCI Factor Crowding Model methodology. Computation for Country factor is adapted for full ESTU sample instead of quintile 1 & 5 samples.

CROWDING SCORES VS FUTURE PERFORMANCE

- Historical relationship between Crowding Scores and subsequent periods' market risk and return.
- Future returns are correlated to Crowding Scores in different ways. Over a 6 month period, Valuation and Pairwise Correlation are negatively correlated with future return but Factor Volatility and Momentum show weakly positive correlation.
- The Crowding Scores showed relatively higher correlation with market risks in the subsequent 12 month period .

Correlation with Future **Return**	0-6m	7-12m	13-18m	19-24m
Valuation	-0.16	-0.36	-0.26	-0.27
PairwiseCorrelation	-0.16	0.10	0.07	0.10
Factor Volatility	0.04	0.12	0.08	0.05
Momentum	0.03	-0.06	-0.02	-0.12
Integrated	-0.09	-0.08	-0.10	-0.22

Correlation with Future **Risk**	0-6m	7-12m	13-18m	19-24m
Valuation	0.11	-0.01	-0.15	-0.26
PairwiseCorrelation	0.37	0.05	-0.22	-0.20
Factor Volatility	0.58	0.22	0.02	0.10
Momentum	0.26	0.33	0.24	0.16
Integrated	0.54	0.27	-0.05	-0.11

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Feb 2000 to April 2018

CROWDING SCORES VS FUTURE PERFORMANCE

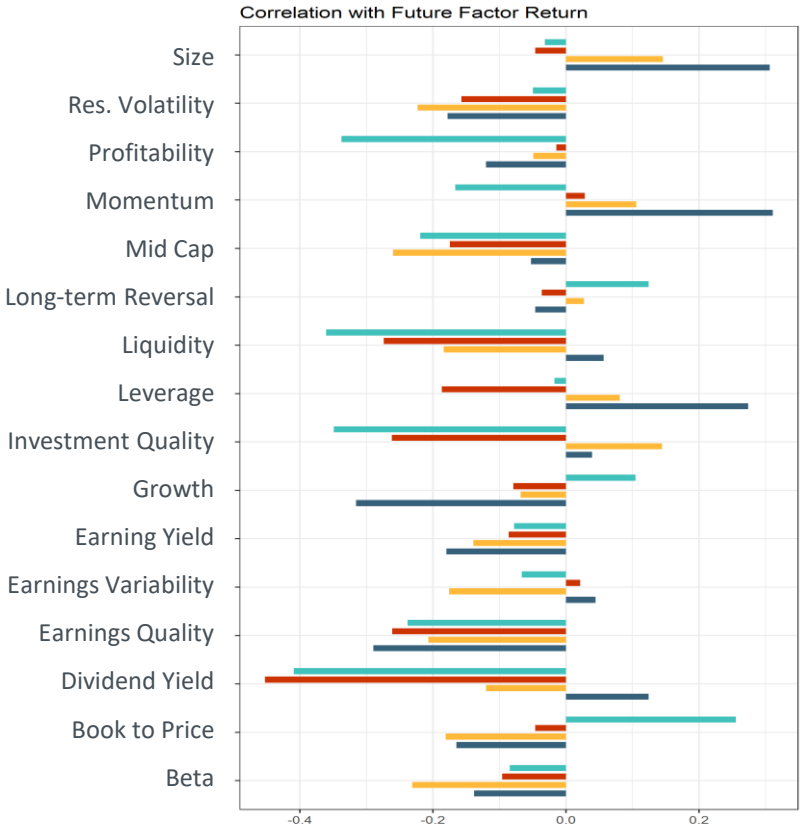
- Positive Integrated Crowding Scores were historically correlated with the bull market up to a certain threshold.
- Low levels of crowding appeared to correlate with mean reverting behavior of the market.

Correlation of Crowding Score with subsequent market performance

	Integrated Crowding Score <=										Integrated Crowding Score >=																
Future Period	-0.9	-0.8	-0.7	-0.6	-0.5	-0.4	-0.3	-0.2	-0.1	0	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1	1.1	1.2	1.3	1.4	1.5	1.6
1 month	-0.365	0.393	0.122	-0.053	-0.174	0.005	0.017	-0.003	0.057	0.061	0.074	0.075	0.065	0.160	0.225	0.193	0.196	0.348	0.371	0.375	0.316	0.423	0.599	0.334	0.353	0.521	-0.067
2 months	-0.008	0.482	0.281	-0.047	-0.120	0.005	-0.063	-0.082	-0.031	-0.027	0.007	-0.020	0.031	0.228	0.273	0.259	0.285	0.514	0.510	0.565	0.550	0.519	0.500	0.225	0.267	0.264	-0.393
3 months	-0.594	0.126	0.174	-0.073	-0.140	-0.062	-0.142	-0.166	-0.085	-0.044	-0.022	-0.022	-0.003	0.257	0.299	0.269	0.338	0.594	0.634	0.652	0.614	0.599	0.442	-0.039	0.125	-0.125	-0.311
4 months	-0.651	0.182	0.177	-0.098	-0.177	-0.081	-0.216	-0.218	-0.159	-0.099	-0.078	-0.057	-0.049	0.189	0.246	0.248	0.369	0.572	0.626	0.654	0.636	0.622	0.276	-0.216	-0.073	-0.189	-0.746
5 months	-0.406	0.276	0.157	-0.148	-0.218	-0.121	-0.261	-0.262	-0.204	-0.144	-0.127	-0.095	-0.096	0.093	0.154	0.175	0.324	0.514	0.593	0.610	0.588	0.595	0.283	-0.165	-0.076	-0.238	-0.168
6 months	-0.384	0.283	0.213	-0.120	-0.223	-0.164	-0.314	-0.319	-0.244	-0.163	-0.163	-0.142	-0.169	-0.007	0.053	0.092	0.264	0.455	0.524	0.533	0.492	0.493	0.146	-0.183	-0.044	-0.218	-0.609

FACTOR CROWDING – INTEGRATED SCORE

- Integrated Crowding showed generally negative correlation to future factor return.



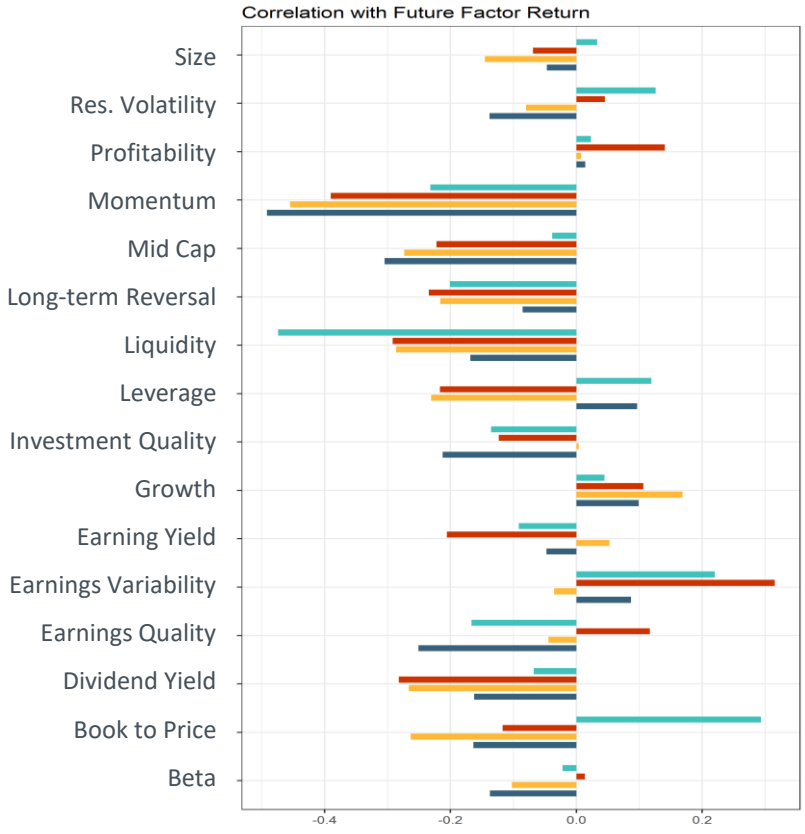


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Feb 2000 to April 2018

FACTOR CROWDING– FACTOR VALUATION

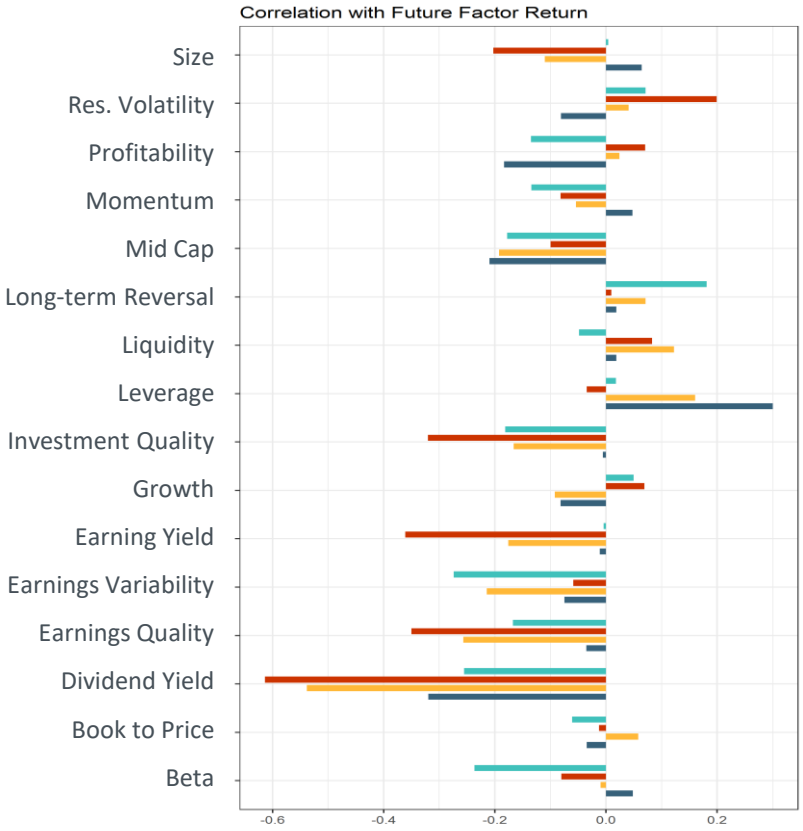
- Valuation is most negatively correlated with Momentum future return.





FACTOR CROWDING– PAIRWISE CORRELATION

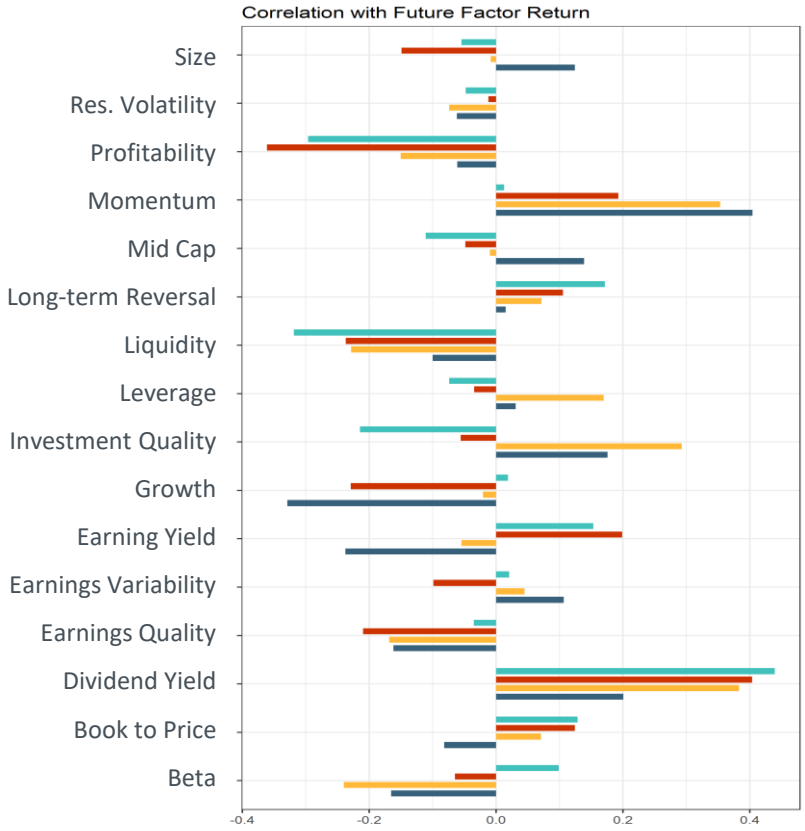
- Pairwise Correlation showed generally negative correlation to future factor return, especially Dividend Yield and Earnings Quality.





FACTOR CROWDING– FACTOR VOLATILITY

- Factor Volatility has positive correlation with Momentum and Dividend Yield's future return.





FACTOR CROWDING– FACTOR MOMENTUM

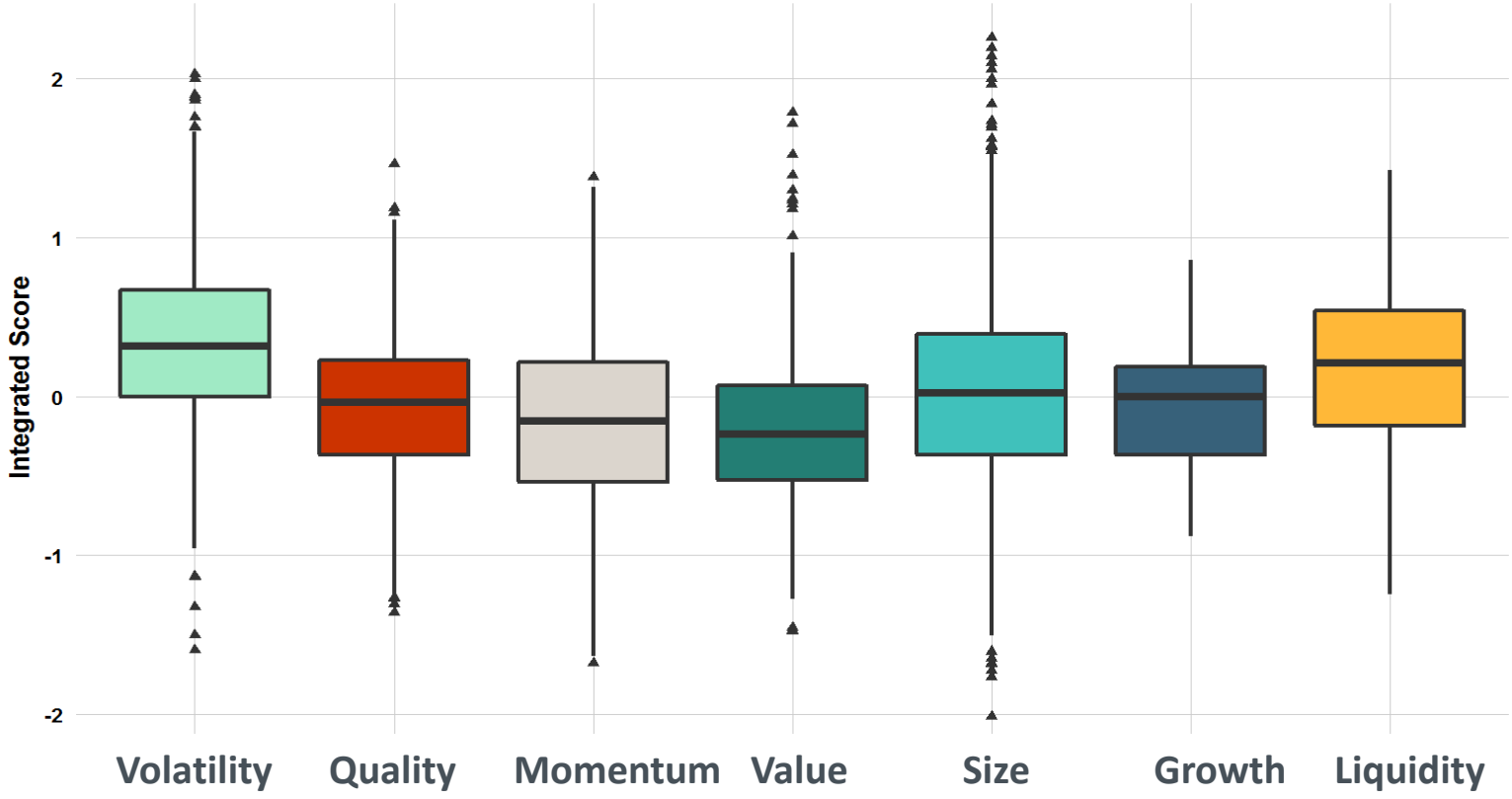
- Factor Momentum showed generally positive correlation to future factor return, especially Size and Liquidity.





CROWDING BY FACTOR TYPE

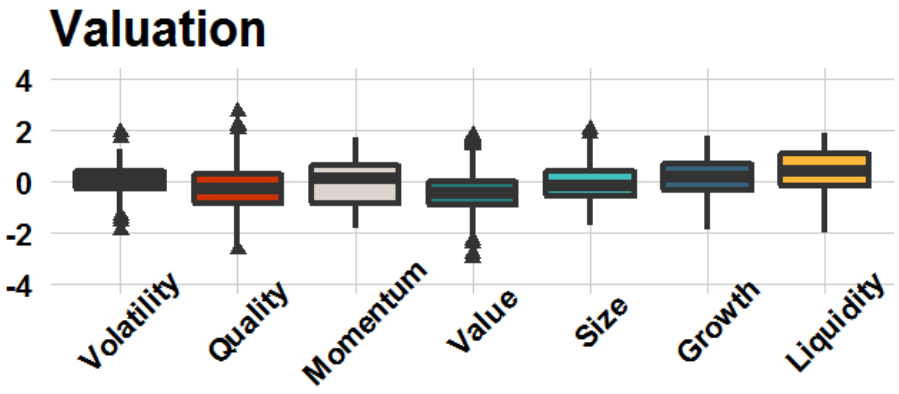
- Box charts showed the Integrated Crowding Score distribution (1st quartile, Median, and 3rd quartile) as well as their max/min outliers.

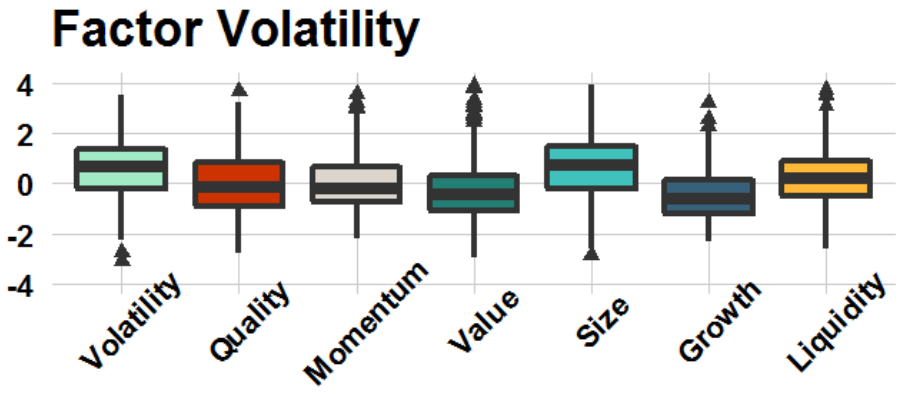


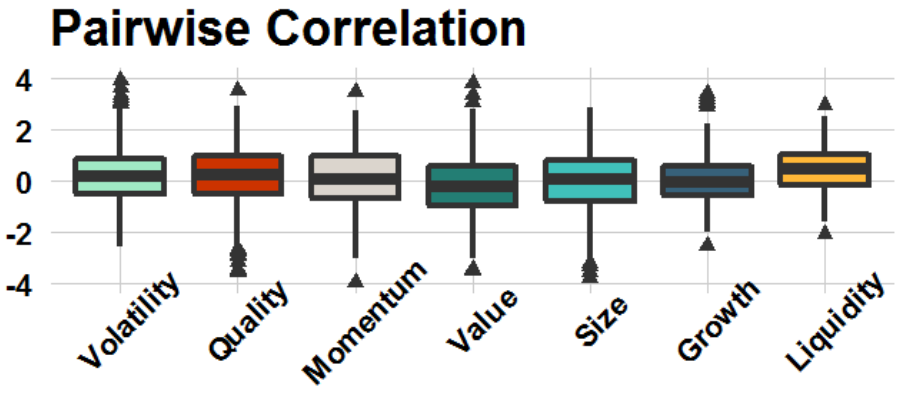
Feb 2000 to April 2018, dividend yield is in "Value" group

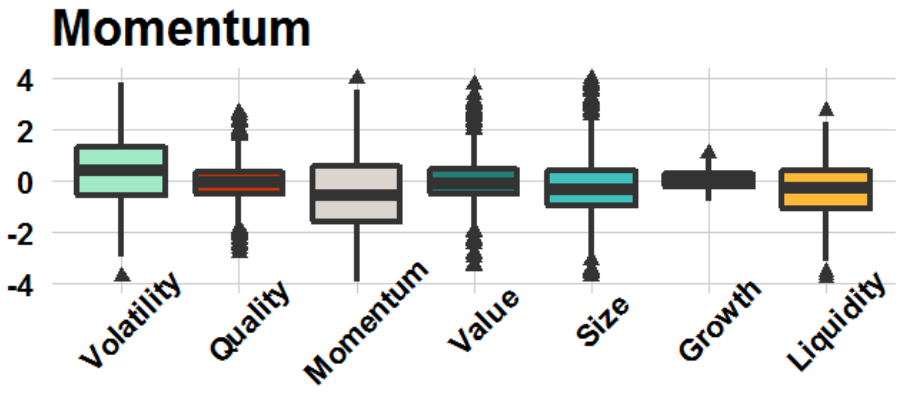
CROWDING BY FACTOR TYPE

- Crowding scores distribution differ greatly by factor classification.







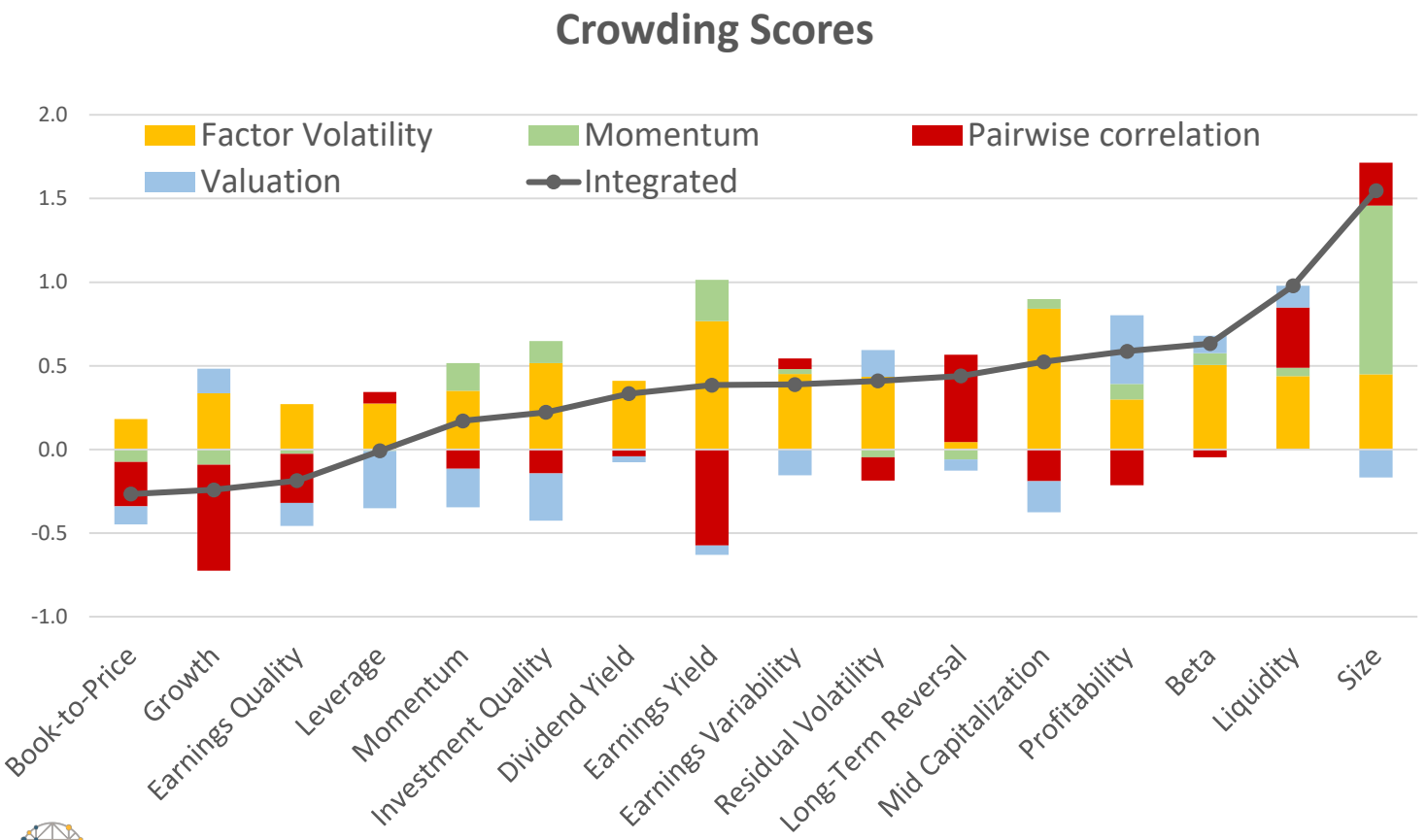


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Feb 2000 to April 2018

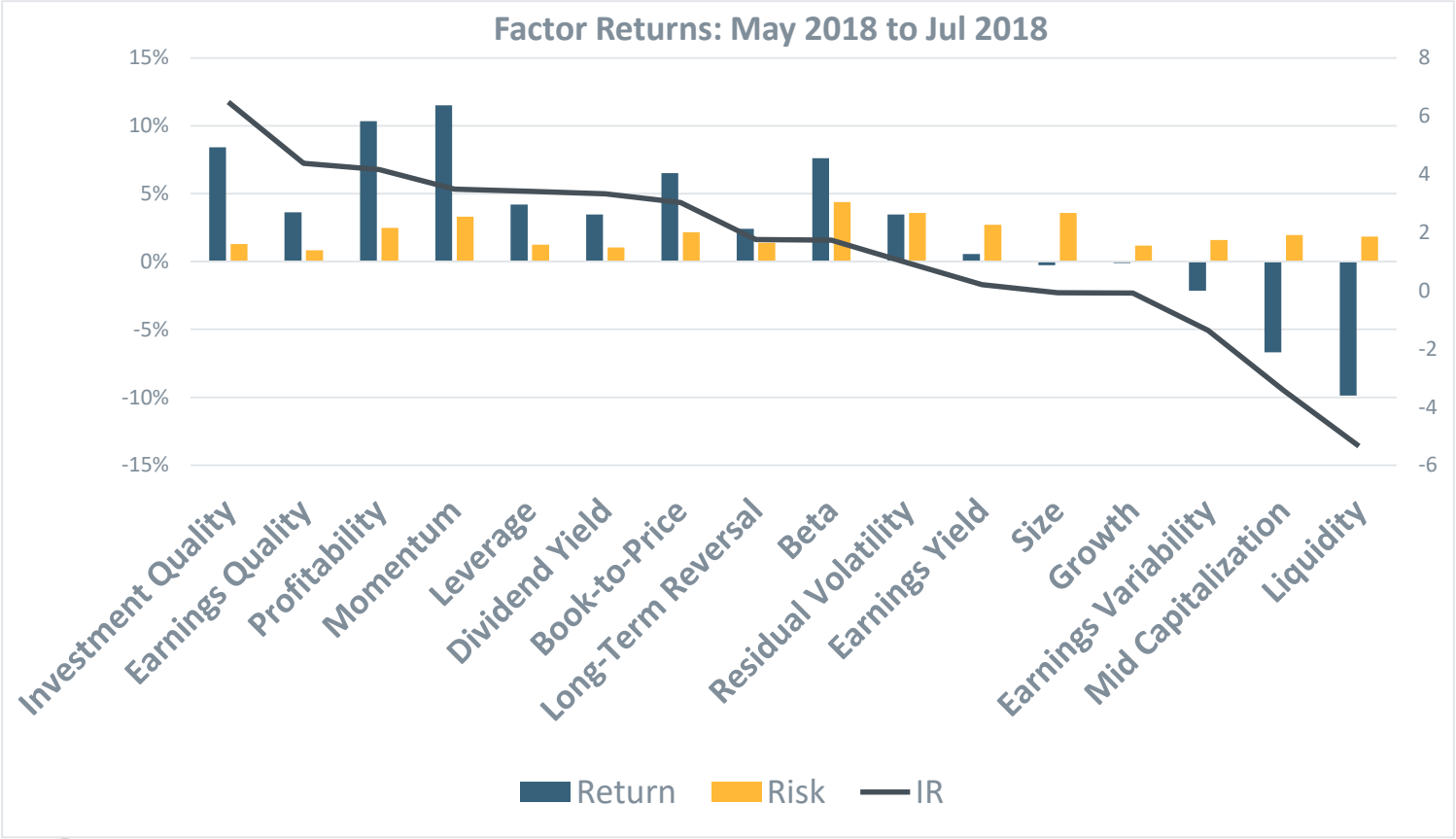
RECENT FACTOR CROWDING

- Size and Liquidity were the most crowded factors on 27 April 2018.



RECENT FACTOR CROWDING

- Both Size and Liquidity displayed weak return and IR in the subsequent months.



A FACTOR FRAMEWORK FOR CHINA

WHY FACTOR STRATEGIES CAN WORK IN CHINA

While we have seen evidence that factor-based strategies could have worked with China A shares, a key question is "Why?" Investors can harness the real power of factors only when they understand why factor strategies could demand long-term compensation of returns – and how these reasons apply to China A shares. The first three reasons are more universal in nature, while the last two reflect more specific influences of "China factors" on market behavior.

- **Systematic risks**: Risks to specific stock traits cannot be diversified away, and factors may have earned excess returns specifically because there was systematic risk attached. For example, the size factor was a case where investors bore higher credit and liquidity risk for the return compensation.

- **Behavioral biases:** Another reason factors may have earned excess returns is because of investor behaviors, such as chasing winners, overreacting or a preference for "local" securities. In China, retail investors sought short-term gains and overpaid for stocks with unrealistic growth projections, whereas patient investors in value and quality dividend stocks were rewarded over the period we examined

- **Institutional constraints:** The third reason China A share factors may have earned excess returns are the frictions and flows that arise from regulatory and industry practices and constraints. For example, many institutions faced short-term pressures, such as monthly performance ranking and quarterly internal asset allocation evaluations. This has led to a preference for higher-risk strategies, creating opportunities for low volatility stocks.

- **Influence of government policies:** Since 2014, China securities regulators have been actively pushing companies to pay higher dividends to shareholders. This has contributed to a positive return premium on dividend-paying stocks in recent years. On the other hand, the lingering perception of state influence, particularly from the so-called "national team" (a market-stabilization fund established during the 2015 crisis), can distort the price discovery process and make the market less efficient.

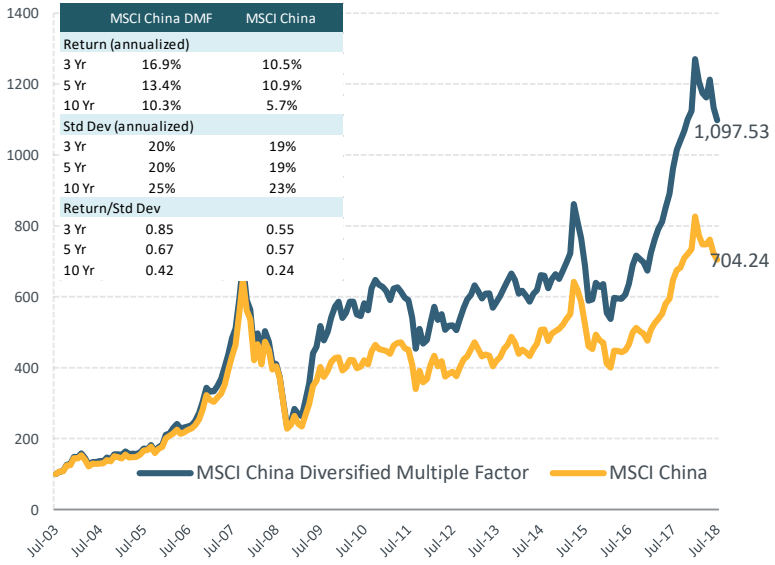
- **Opening of China A Shares market:** One direct consequence of the opening of the China A shares market and their partial inclusion in the MSCI Emerging Markets Index is greater institutional participation in the market. Institutional investors historically have been more attuned to stocks with attractive valuation, dividend payout and stronger fundamentals, which may help explain why yield, value and quality factor indexes performed better during the past two to three years.

LOOKING INTO CHINA THROUGH MSCI FACTOR LENS

- The importance of stock selection performance in A shares market will grow as China's weight within the MSCI EM Index steadily increases
- For active managers this offers an immense opportunity to build a track record and market credibility to attract current investors, but also to prepare for the growing number of investors outside China that will seek China investment skill.
- Factor-based investing has worked well historically, as illustrated by MSCI China DMF and MSCI China A DMF indexes*
- Managers can start to build track record on MSCI China/China A, while using the DMF indexes as internal secondary benchmark and our models for better analysis
- The important enhancements we have made on our Factor Models now offer active managers a richer breadth of factor insights

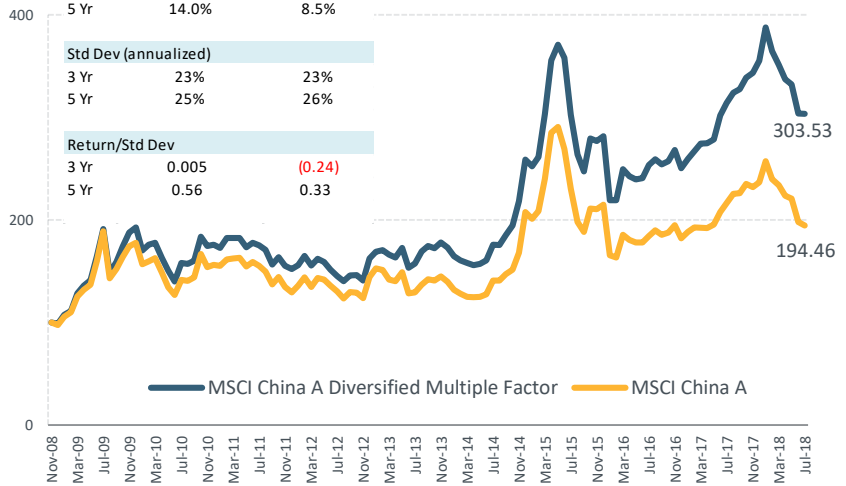
MSCI China DMF

	MSCI China DMF	MSCI China
Return (annualized)		
3 Yr	16.9%	10.5%
5 Yr	13.4%	10.9%
10 Yr	10.3%	5.7%
Std Dev (annualized)		
3 Yr	20%	19%
5 Yr	20%	19%
10 Yr	25%	23%
Return/Std Dev		
3 Yr	0.85	0.55
5 Yr	0.67	0.57
10 Yr	0.42	0.24



MSCI China A DMF

	MSCI China A DMF	MSCI China A
Return (annualized)		
3 Yr	0.1%	-5.5%
5 Yr	14.0%	8.5%
Std Dev (annualized)		
3 Yr	23%	23%
5 Yr	25%	26%
Return/Std Dev		
3 Yr	0.005	(0.24)
5 Yr	0.56	0.33

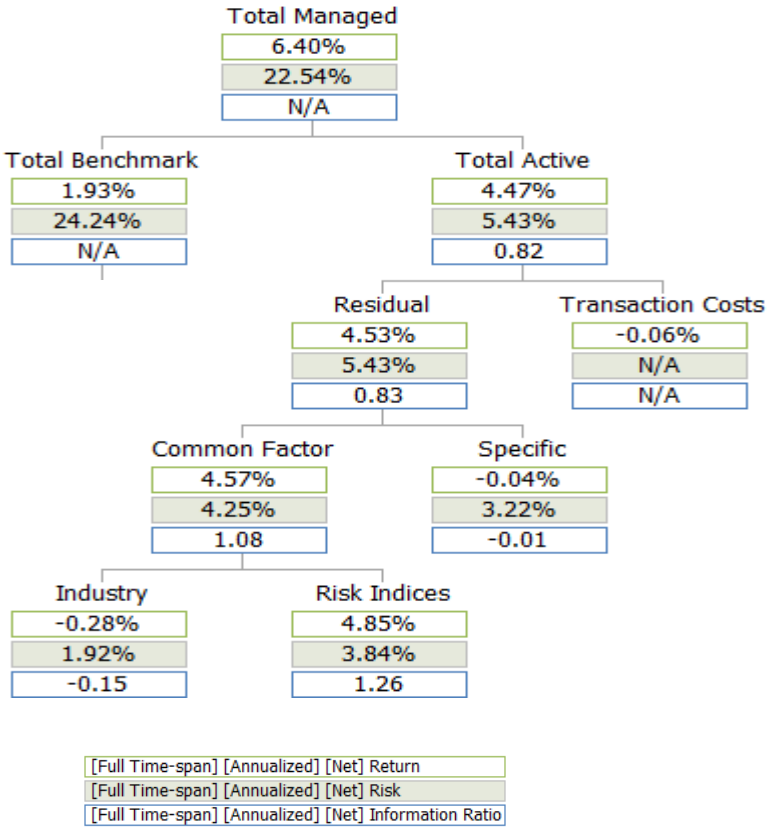


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*DMF (Diversified Multiple Factor) index aims to maximize exposure to four factors – Value, Momentum, Quality and Low Size -- while maintaining a risk profile similar to that of the underlying parent index. Charts and tables are based on monthly gross total return USD. MSCI China A DMF is simulated, not launched yet.

PERFORMANCE ANALYSIS WITH CNLT

- Outperformed 4.47%.
- Liquidity and Earning Yield contributed almost 2.96% and 1.56%.





Benchmark: MSCI China A ,Period: Nov 2009 to April 2018 Monthly

SIMULATING FACTOR STRATEGY WITH CNLT

Case 1 : Construct a multi factor portfolio using CNLT

Case 2 : Enhance the portfolio above with dynamic factors allocation based on Integrated Crowding Scores.

In both cases, we compute simulated performance results against MSCI China A index for the period Nov 2009 to April 2018, assuming monthly rebalancing.

CASE 1: MULTI FACTOR

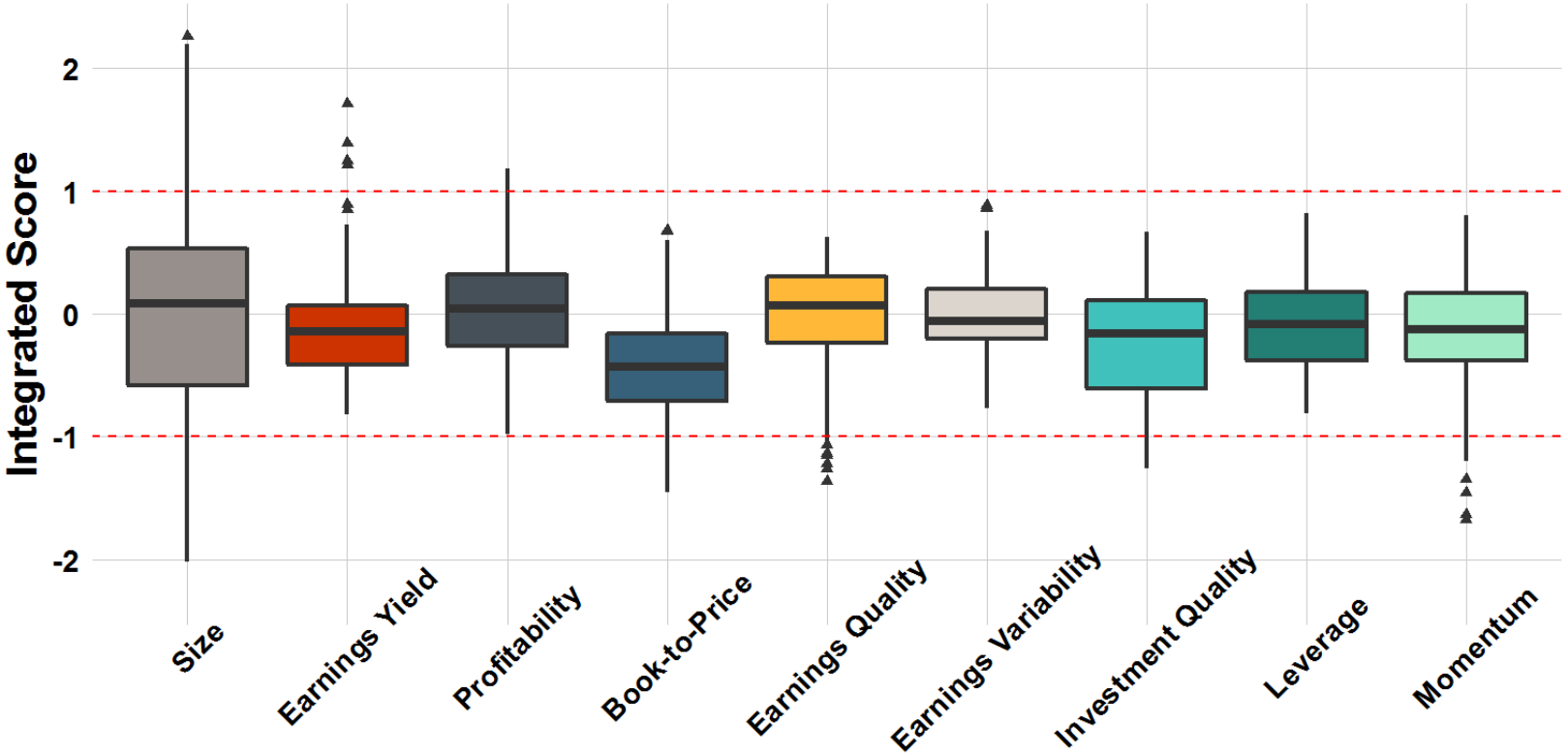
Objective: Maximize the portfolio exposure to a diversified set of CNLT factor groups while maintaining market risk similar to the MSCI China A

*Expected Return = 0.25 * Momentum + 0.25 * Low Size + 0.25 * Value + 0.25 * Quality*

Constraint	Min	Max
Total Risk		Benchmark Risk
Beta	b-0.25	b+0.25
Dividend Yield	b-0.25	b+0.25
Growth	b-0.25	b+0.25
Liquidity	b-0.25	b+0.25
Long-Term Reversal	b-0.25	b+0.25
Liquidity	b-0.25	b+0.25
Mid Capitalization	b-0.25	b+0.25
Residual Volatility	b-0.25	b+0.25
Size	b-1	b+1
Sector	b-5	b+5
Asset Level	MAX(Bmk Weight - 0.02,0)	MIN(Bmk Weight * 10,Bmk Weight + 0.02)

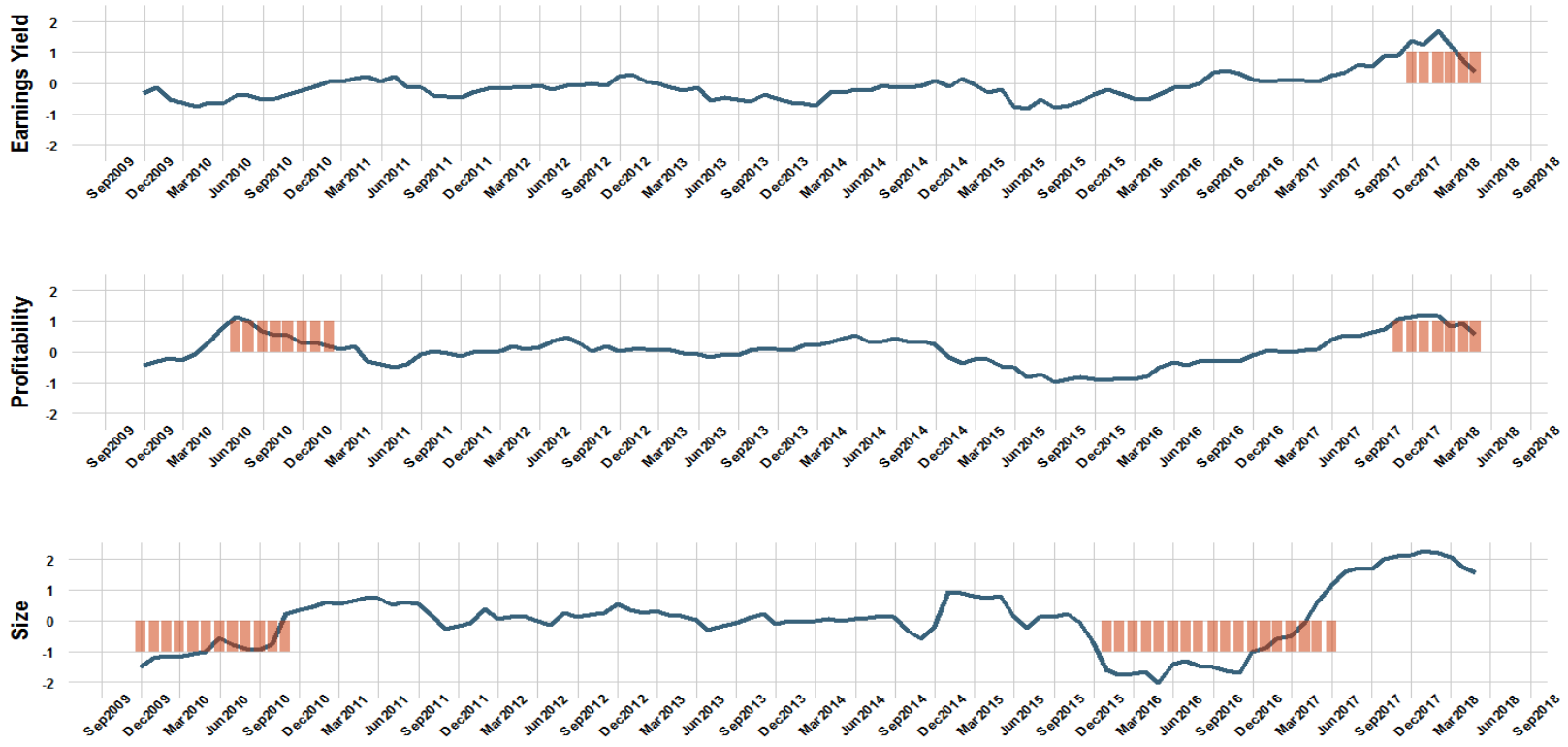
CASE 2: DYNAMIC FACTOR ALLOCATION

- Earning Yield and Profitability, Low Size have shown higher levels of crowding during back testing period



CASE 2: DYNAMIC FACTOR ALLOCATION

- Factor weights for Earnings Yield, Profitability and Low Size are reduced to zero for six consecutive months in the Expected Return when their respective Integrated Score exceeds the threshold of 1.



PERFORMANCE COMPARISON

Portfolio	Benchmark	Annualized in CNY (%)				
		Portfolio	Benchmark	Active	Tracking Error	Information Ratio
MSCI China A DMF	MSCI China A	6.4	1.93	4.47	5.43	0.82
Case 1: CNLT Multi-factor	MSCI China A	7.16	1.93	5.23	4.58	1.14
Case 2: CNLT Multi-factor with dynamic factor allocation	MSCI China A	8.37	1.93	6.44	4.79	**1.35**

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Period: Nov 2009 to April 2018 Monthly

CONCLUSION

- New style factor structure in CNLT increases the dimension of factor study.
- Crowding can be quantified by Crowding Scores which enables investors to monitor and examine potential downside risk in their strategies.
- A complete factor framework for China A shares market can be built with new MSCI China Equity Model and MSCI China A DMF index.



MSCI ANALYTICS PLATFORM

The Power of Connection.

Arman Sahovic
Senior Associate l Client Coverage

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MSCI中国固定收益市场模型

回顾及个案分析

Tommy Ip

Vice President l Client Coverage

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议程

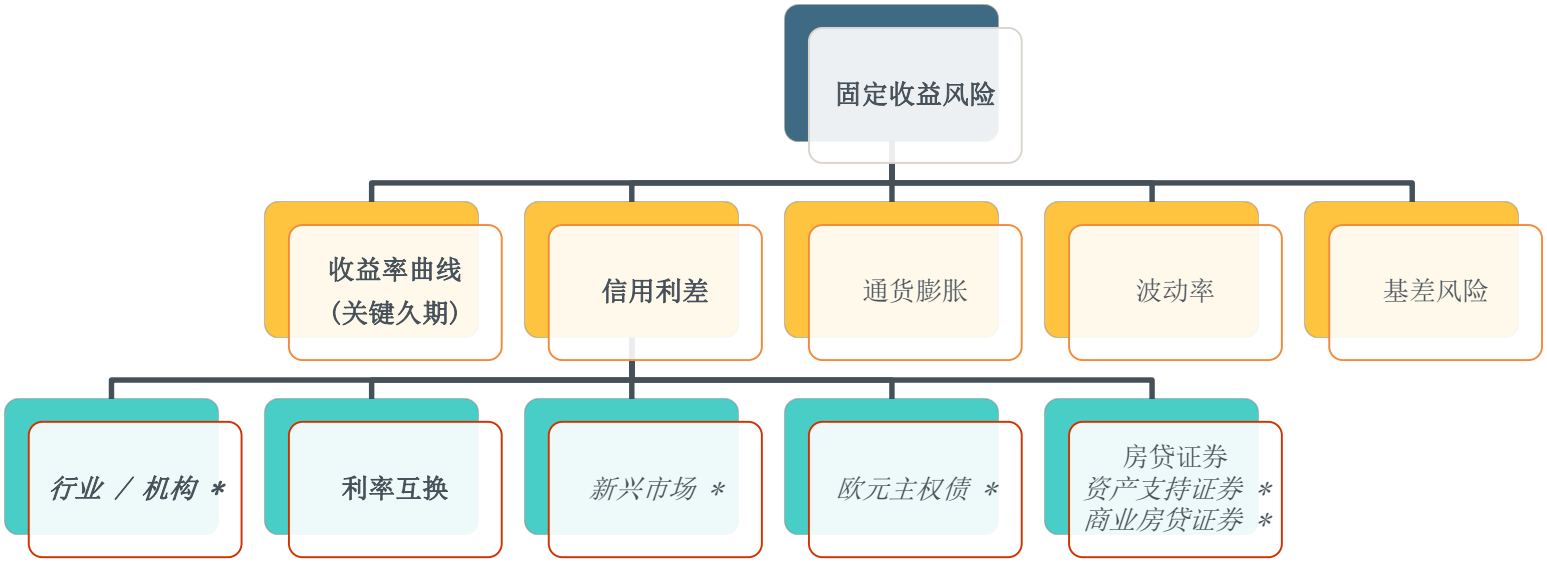
01. 模型概观及期限曲线

02. 信用利差

03. 高收益债券个案分析

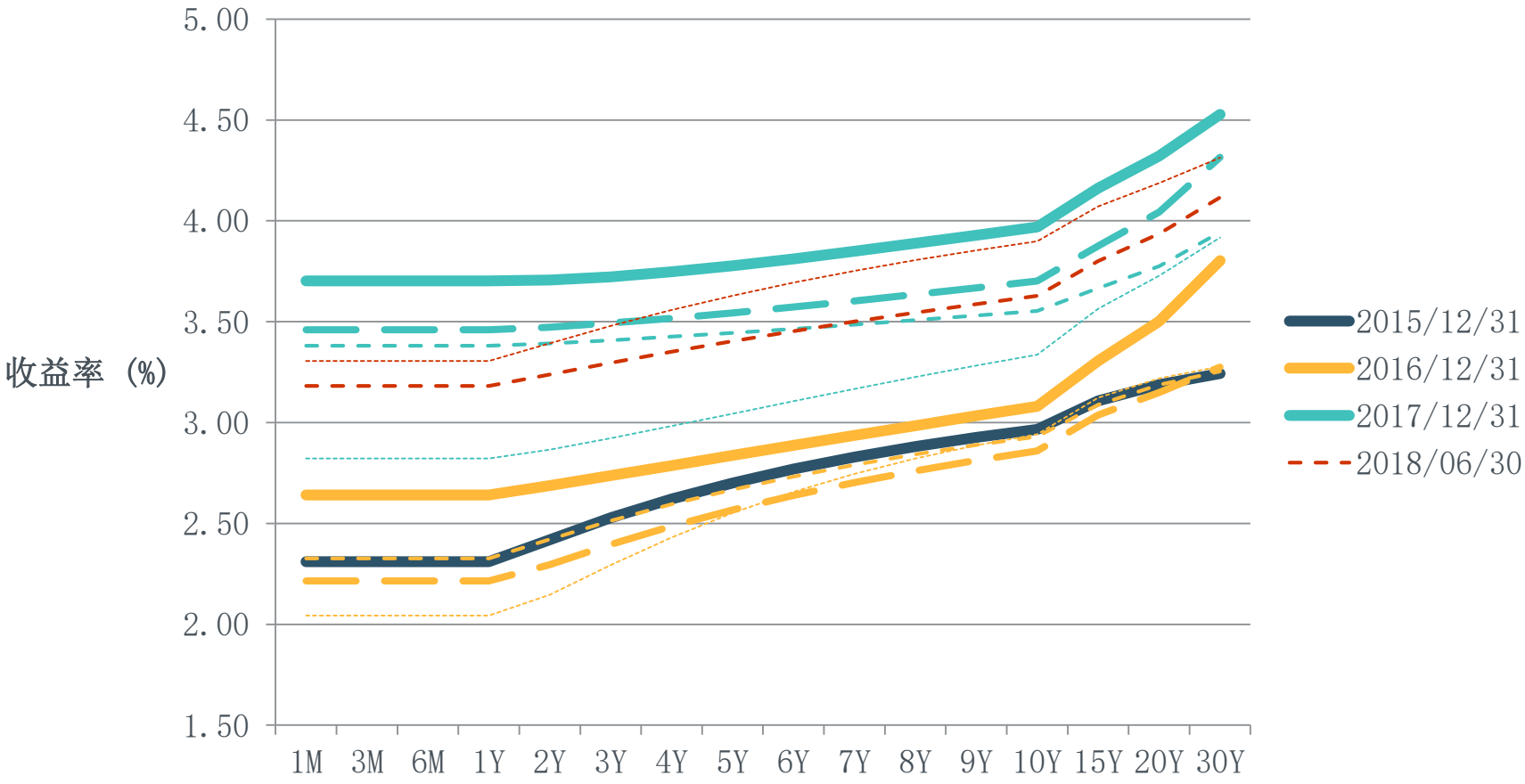
01. 模型概观及期限曲线

模型因子结构



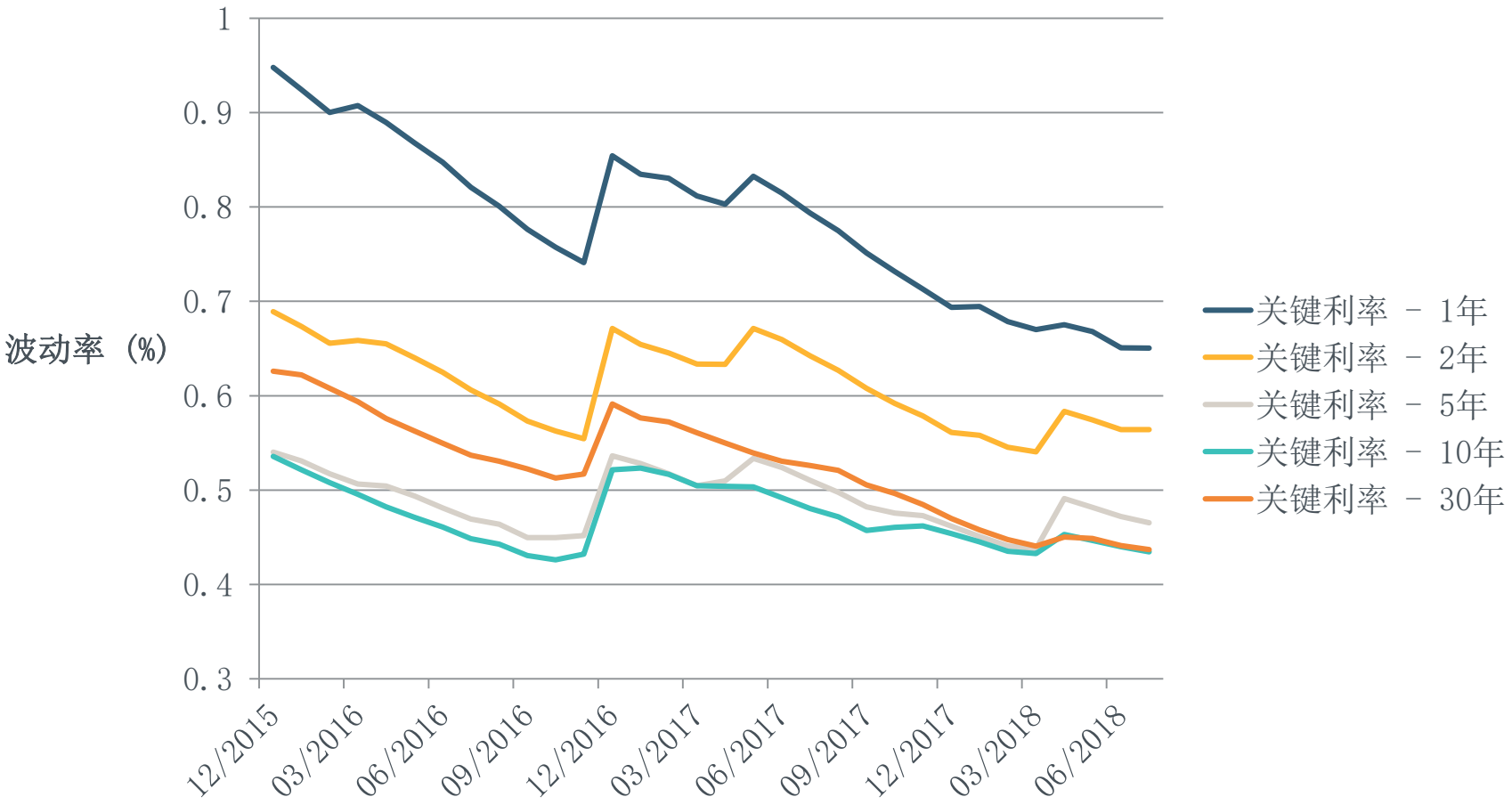
* "DTS"久期x 信用利差模型

收益率曲线

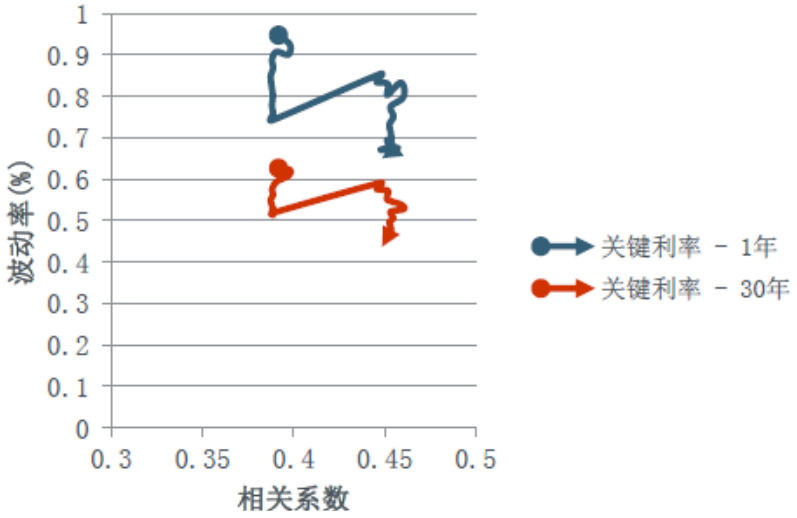


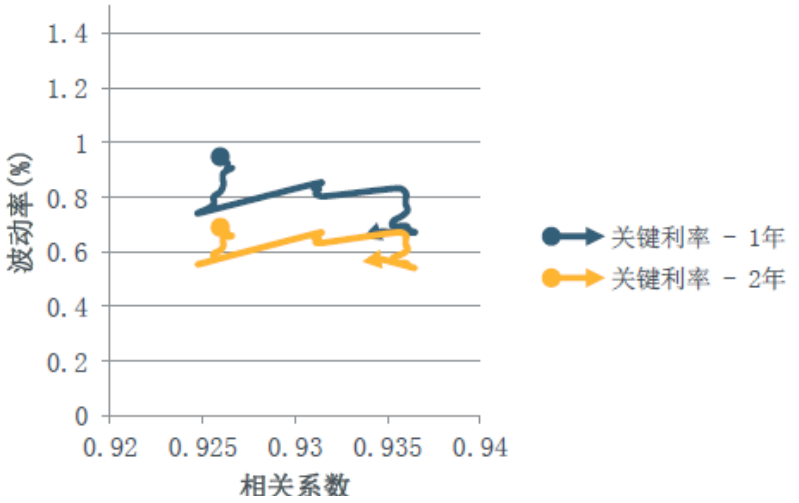
来源：MSCI BarraOne系统

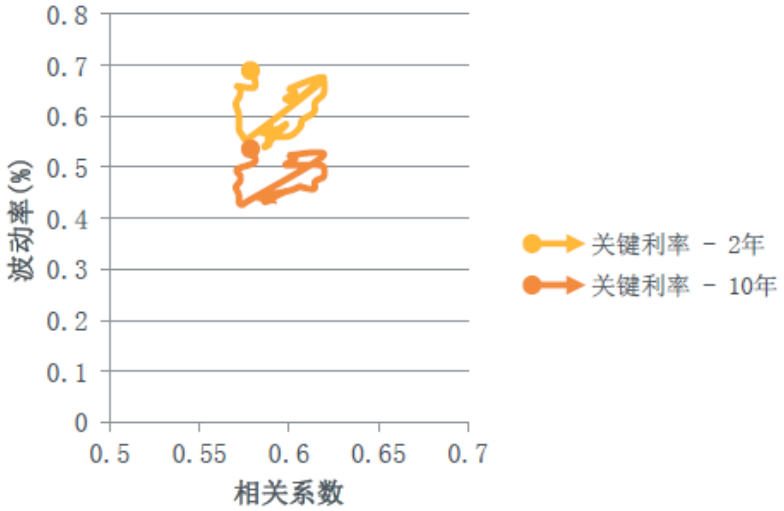
关键利率因子

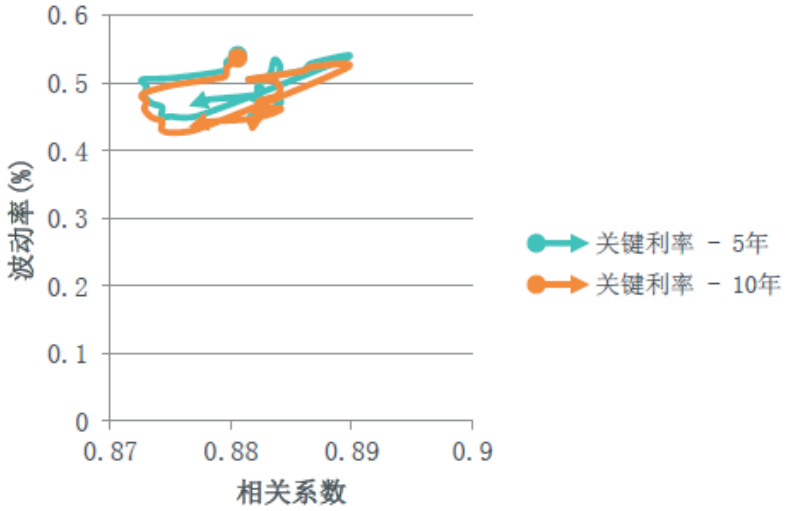

波动率（%）
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
12/2015
03/2016
06/2016
09/2016
12/2016
03/2017
06/2017
09/2017
12/2017
03/2018
06/2018
关键利率 - 1年
关键利率 - 2年
关键利率 - 5年
关键利率 - 10年
关键利率 - 30年

关键利率波动率和相关系数







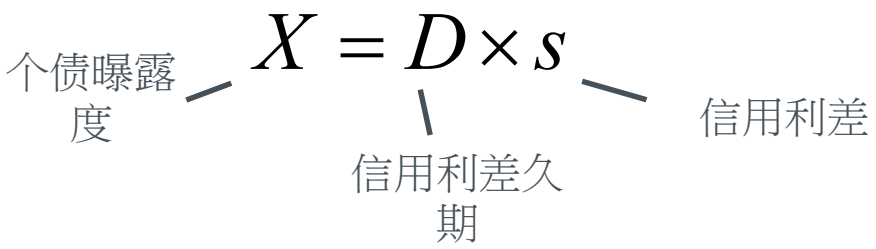


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02. 信用利差

DTS模型

- 信用风险曝露度指标 = 久期 x 信用利差 “Duration Times Spread”



- 信用利差越宽 = 风险曝露度越高
- 因子反映*相对*信用利差改变

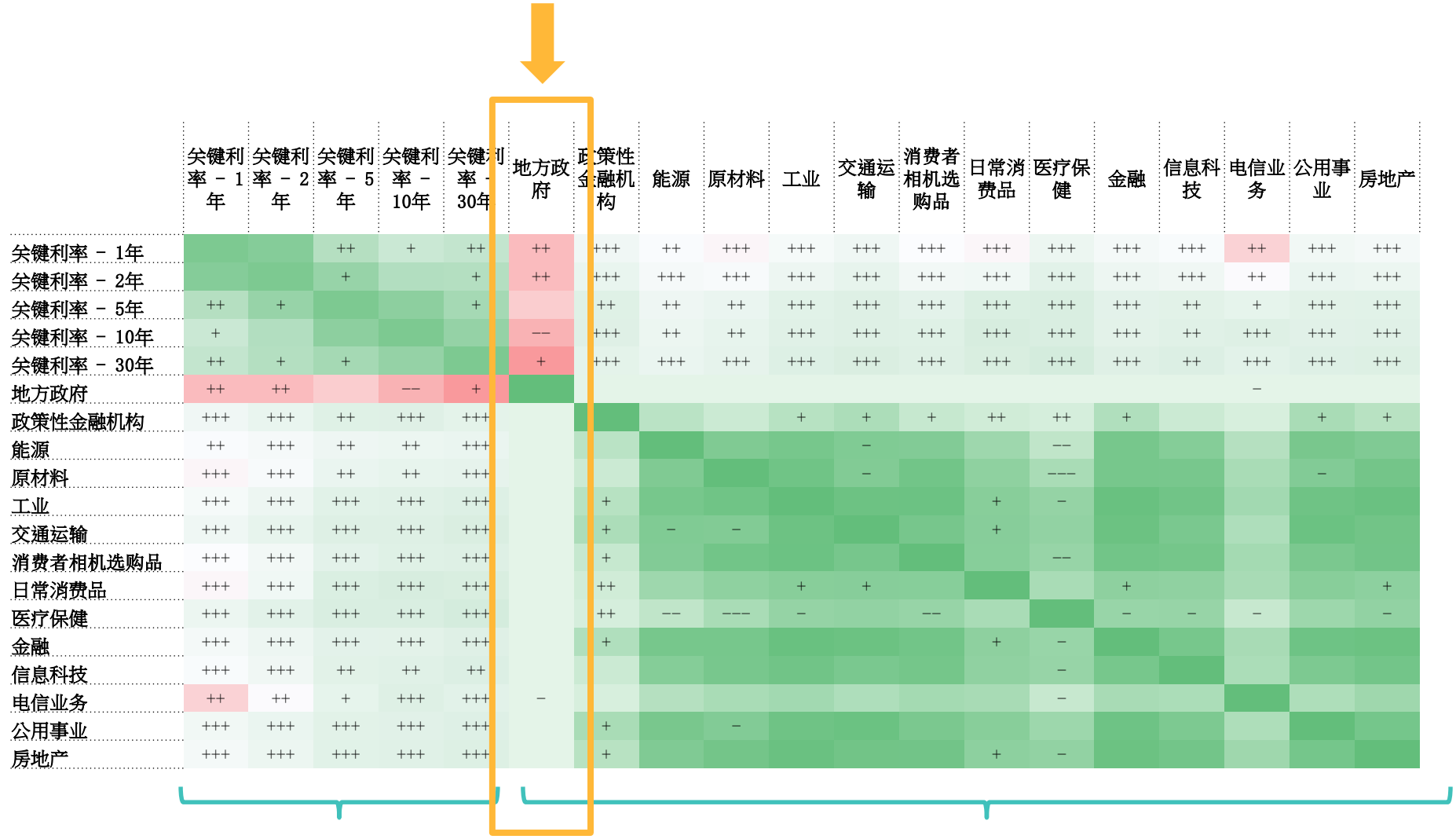
信用利差因子

大类分组	子类别因子数量
地方政府	2
政策性金融机构	1
能源	2
原材料	4
工业	4
交通运输	5
消费者相机选购品	5
日常消费品	2
医疗保健	1
金融	4
信息科技	2
电信业务	1
公用事业	4
房地产	1

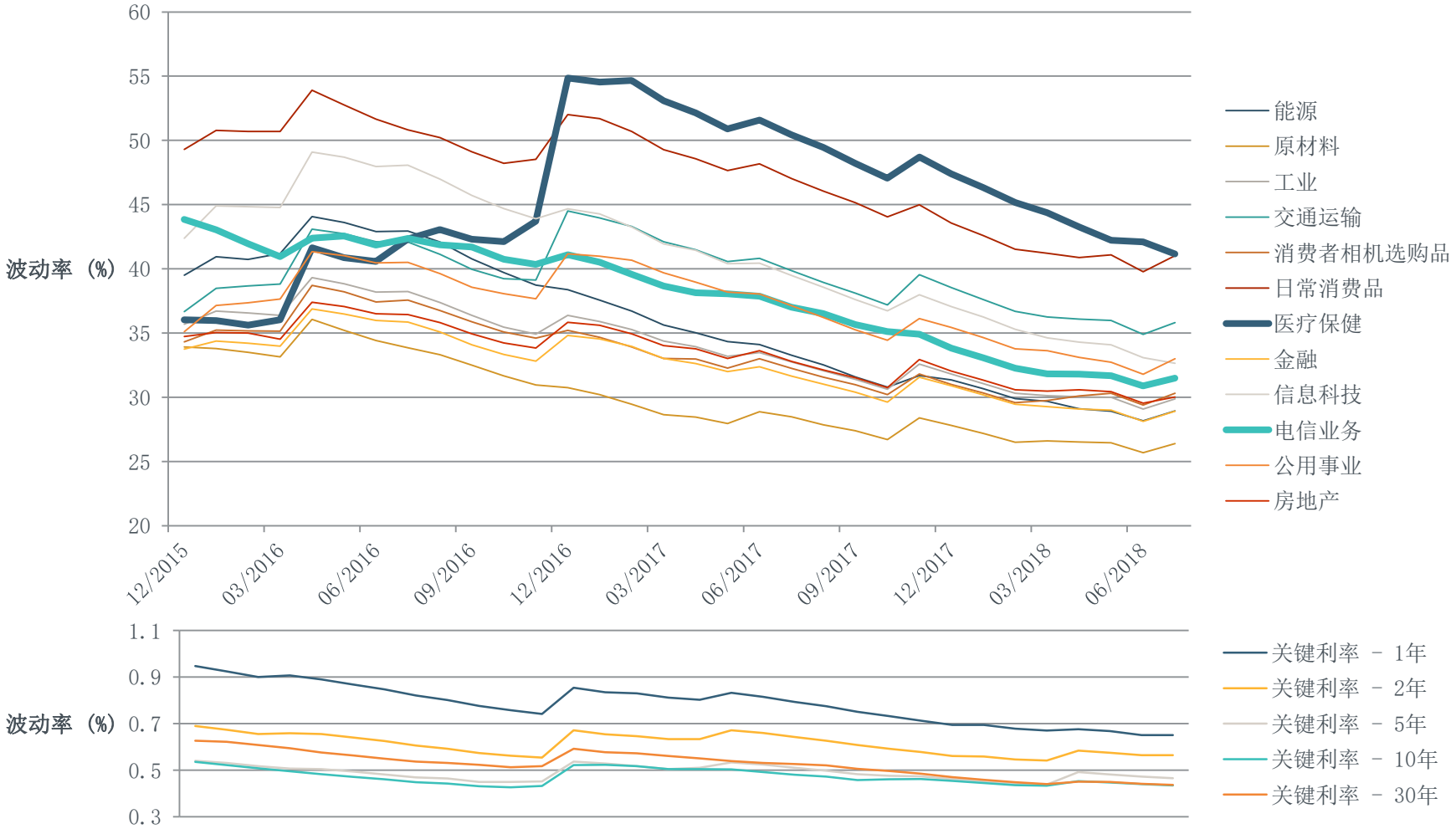
MSCI 全球行业分类标准 (GICS)

来源：MSCI中国固定收益市场模型

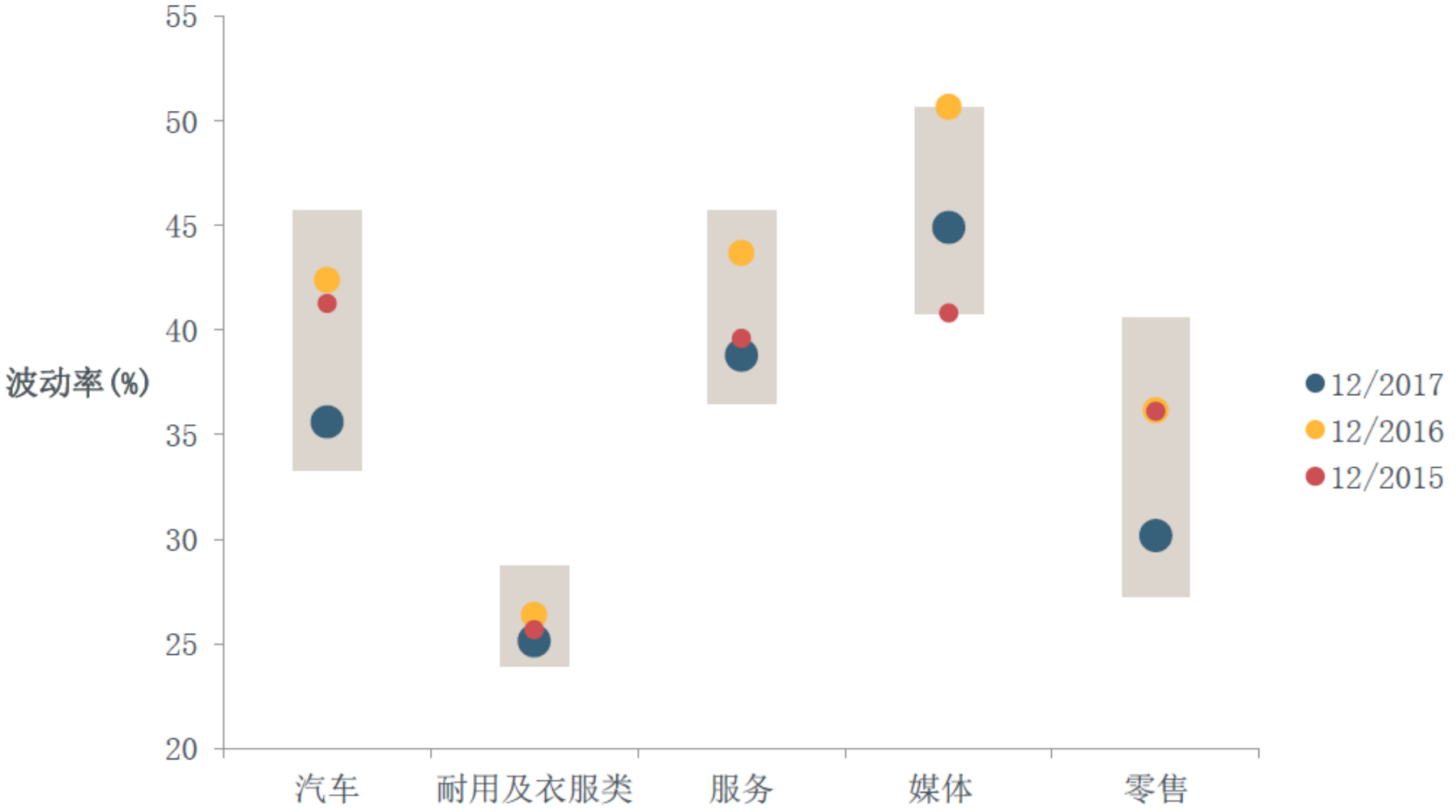
信用利差相关系数


关键利率 – 1年
关键利率 – 2年
关键利率 – 5年
关键利率 – 10年
关键利率 – 30年
地方政府
政策性金融机构
能源
原材料
工业
交通运输
消费者相机选购品
日常消费品
医疗保健
金融
信息科技
电信业务
公用事业
房地产
关键利率 – 1年 ++ + ++ ++ +++ ++ +++ +++ +++ +++ +++ +++ +++ +++ ++ +++ +++
关键利率 – 2年 + + ++ +++ +++ +++ +++ +++ +++ +++ +++ +++ +++ +++ ++ +++ +++
关键利率 – 5年 ++ + + ++ ++ ++ +++ +++ +++ +++ +++ +++ +++ ++ + +++ +++
关键利率 – 10年 + −− +++ ++ ++ +++ +++ +++ +++ +++ +++ +++ ++ +++ +++ +++
关键利率 – 30年 ++ + + + +++ +++ +++ +++ +++ +++ +++ +++ +++ ++ +++ +++ +++
地方政府 ++ ++ −− + − +++ ++ ++ + + + ++ ++ + + + − + +
政策性金融机构 +++ +++ ++ +++ +++ + − − + + + ++ ++ + + + +
能源 ++ +++ ++ ++ +++ + − − − −− −−− + − − − − −
原材料 +++ +++ ++ ++ +++ + − − − −− −−− + − − − − −
工业 +++ +++ +++ +++ +++ + − − + + + + − + − − − −
交通运输 +++ +++ +++ +++ +++ + − − + + + + − + − − − −
消费者相机选购品 +++ +++ +++ +++ +++ + − − + + + −− − − − − −
日常消费品 +++ +++ +++ +++ +++ ++ − − + + + + − − − − − +
医疗保健 +++ +++ +++ +++ +++ ++ −− −−− − −− − − − − − − −
金融 +++ +++ +++ +++ +++ + − − + + + + − − − − − −
信息科技 +++ +++ ++ ++ ++ − − − + + + + − − − − − −
电信业务 ++ ++ + +++ +++ − − − + + + + − − − − − −
公用事业 +++ +++ +++ +++ +++ + − − + + + + − − − − − −
房地产 +++ +++ +++ +++ +++ + − − + + + + − − − − − −
关键利率
信用利差

信用利差波动率

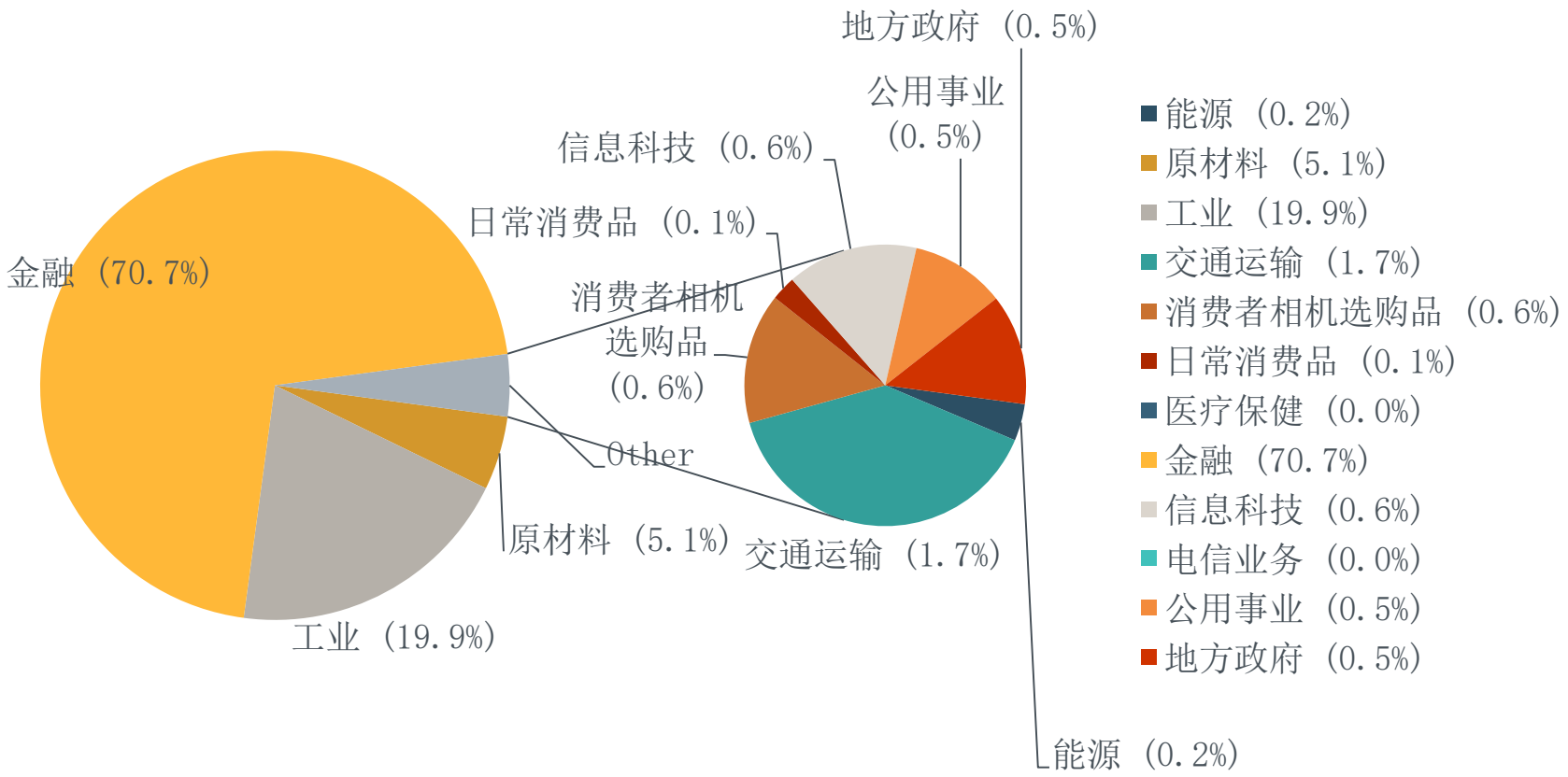

波动率（%）
60
55
50
45
40
35
30
25
20
12/2015
03/2016
06/2016
09/2016
12/2016
03/2017
06/2017
09/2017
12/2017
03/2018
06/2018
能源
原材料
工业
交通运输
消费者相机选购品
日常消费品
医疗保健
金融
信息科技
电信业务
公用事业
房地产
波动率（%）
1.1
0.9
0.7
0.5
0.3
关键利率 - 1年
关键利率 - 2年
关键利率 - 5年
关键利率 - 10年
关键利率 - 30年

消费者相机选购品


55
50
45
40
35
30
25
20
波动率(%)
汽车
耐用及衣服类
服务
媒体
零售
12/2017
12/2016
12/2015

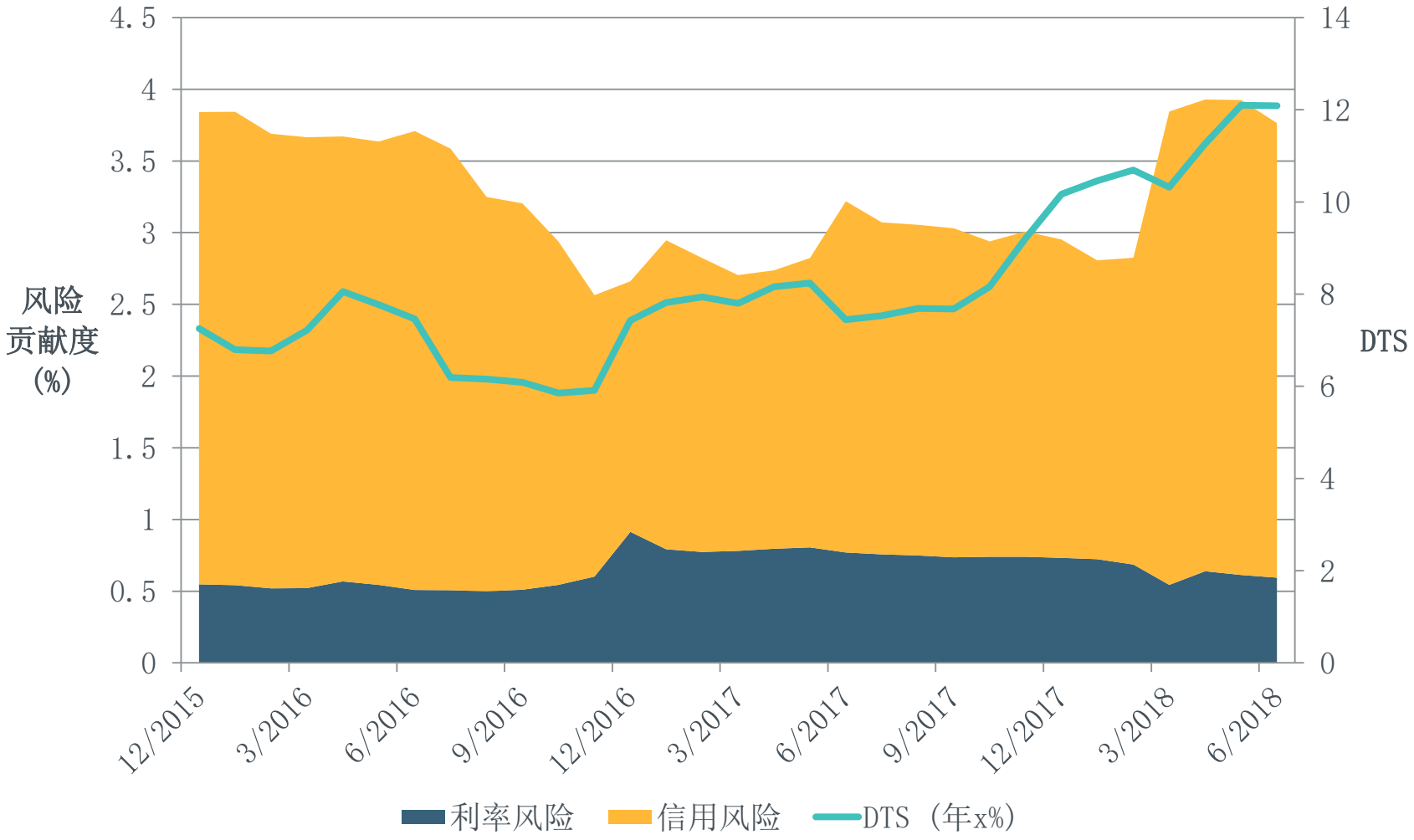
03. 高收益债券个案分析

高收益债券指数

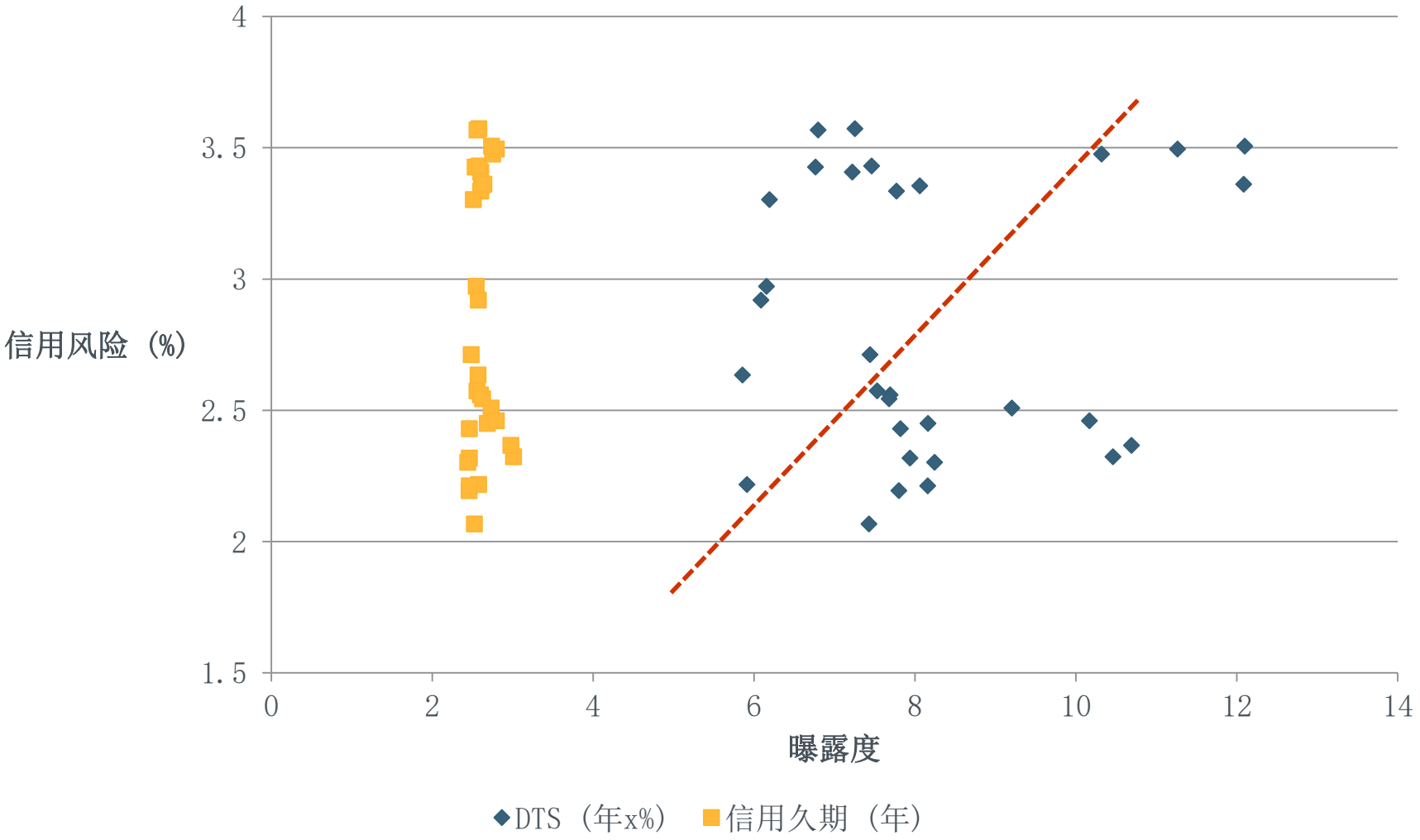


来源：中国在岸高收益债券指数

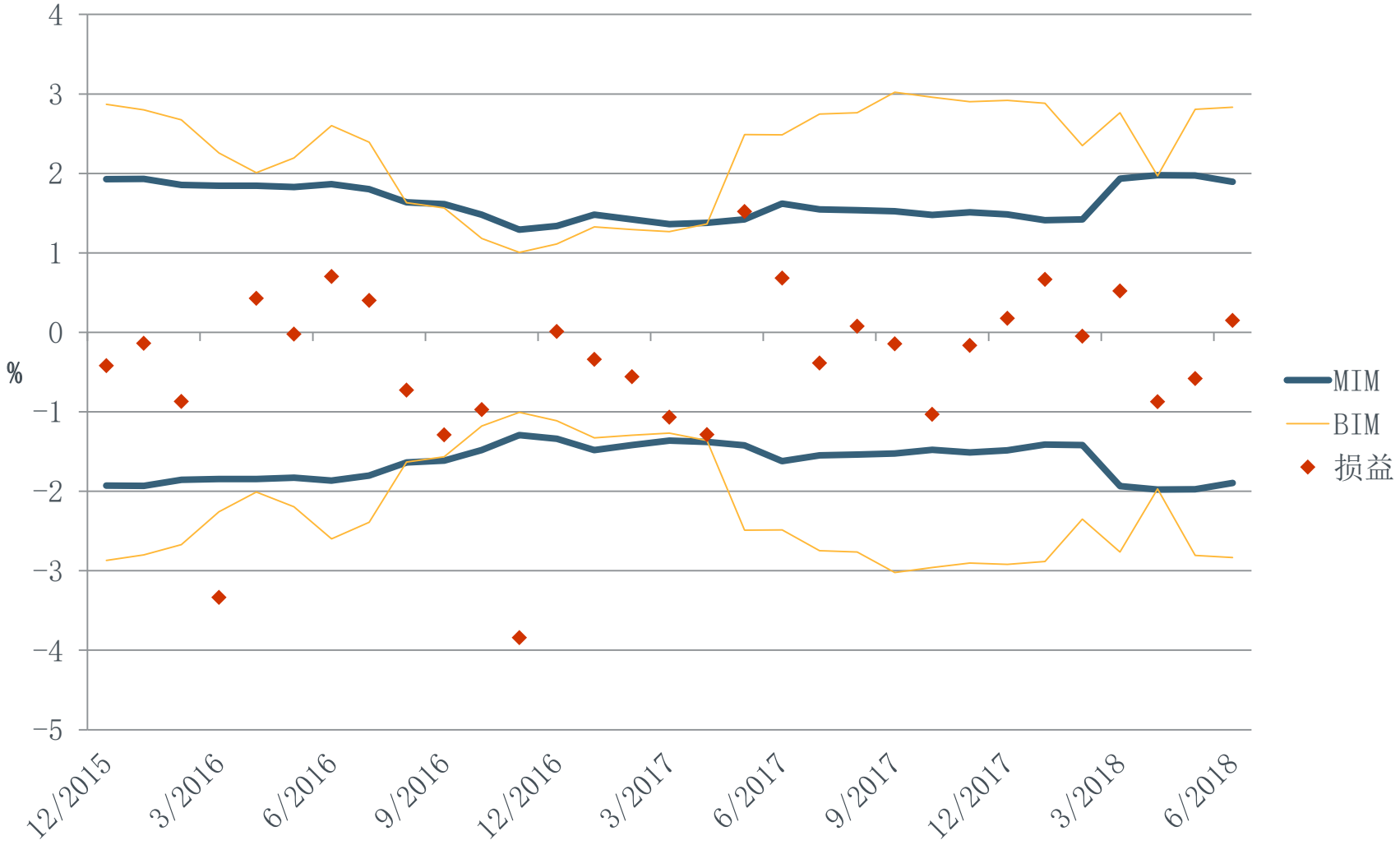
风险拆解


风险贡献度(%)
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
14
12
10
8
6
4
2
0
DTS
12/2015
3/2016
6/2016
9/2016
12/2016
3/2017
6/2017
9/2017
12/2017
3/2018
6/2018
利率风险
信用风险
DTS（年x%）

曝露度指标

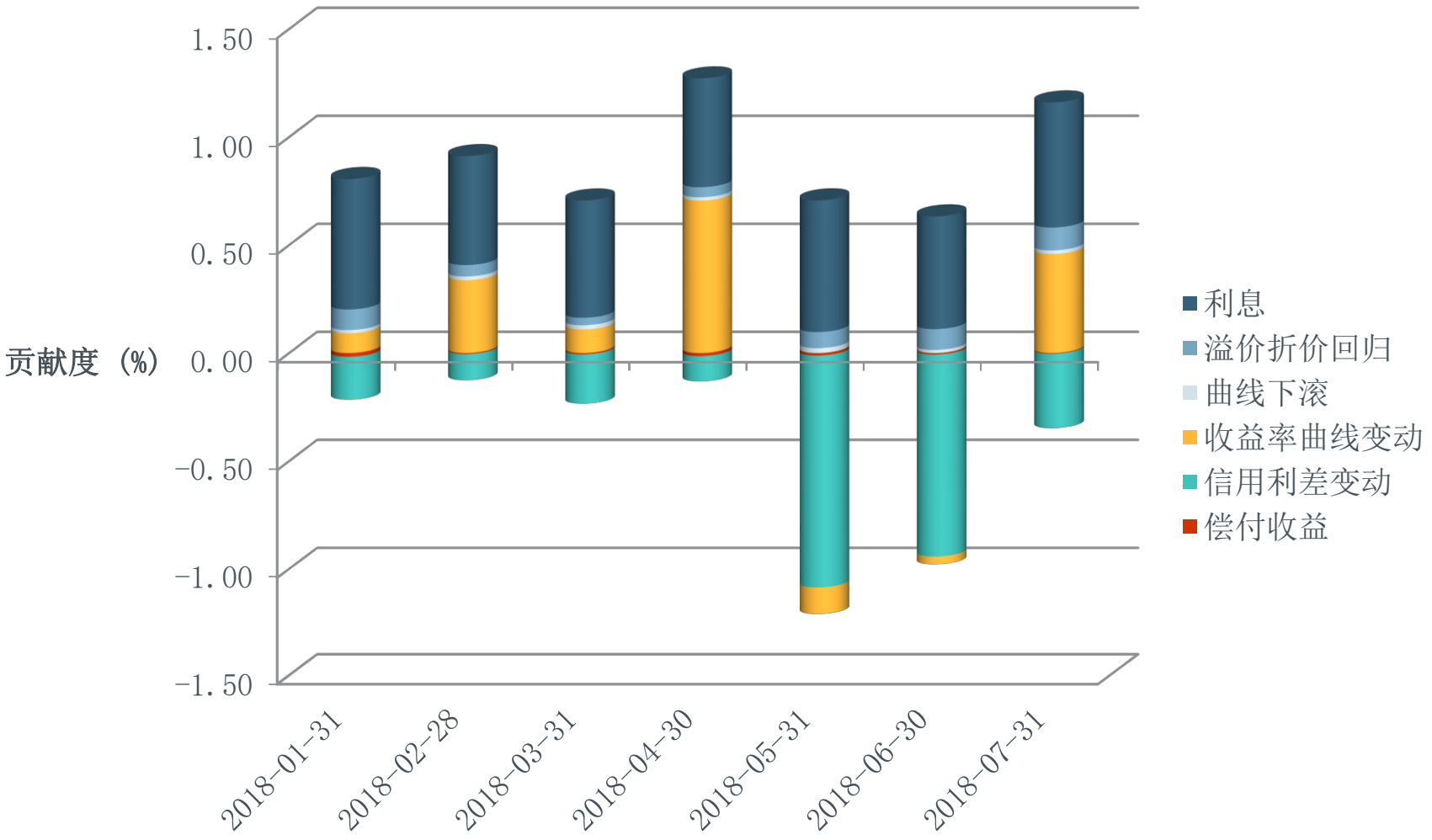


模型验证



来源：中国在岸高收益债券指数

收益对应风险?



来源：中国在岸高收益债券指数

Q&A

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